Examples

Example 1.1

How much charge is represented by 4,600 electrons?

Practice Problem 1.1

Calculate the amount of charge represented by 10 billion protons.

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Examples

Example 1.2

The total charge entering a terminal is given by

$$q(t) = 5t \sin(4\pi t) \text{ mC}.$$

Calculate the current at $t = 0.5$ s.

Practice Problem 1.2

If in Example 1.2,

$$q(t) = \left(20 - 15t - 10e^{-3t}\right) \text{ mC},$$

find the current at $t = 1.0$ s.

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Examples

Example 1.3

Determine the total charge entering a terminal between

$$t = 1\,\text{s} \quad \text{and} \quad t = 2\,\text{s}$$

if the current passing the terminal is

$$i = (3t^2 - t)\,\text{A}$$

Practice Problem 1.3

The current flowing through an element is

$$i = \begin{cases} 8\,\text{A}, & 0 < t < 1 \\ 8t^2\,\text{A}, & t > 1 \end{cases}$$

Calculate the charge entering the element from

$$t = 0 \quad \text{to} \quad t = 2\,\text{s}$$

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Power and Energy

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

Energy is the capacity to do work, measured in joules (J).

The electric power utility companies measure energy in watt-hours (Wh), where

$$1\ \text{Wh} = 3{,}600\ \text{J}$$

Example 1.4

An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb.

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Power and Energy Examples

Practice Problem 1.4

To move charge q from point b to point a requires 100 J.

Find the voltage drop v_{ab} (the voltage at a positive with respect to b) if:

(a) $q = 5\,\mathrm{C}$

(b) $q = -10\,\mathrm{C}$

Example 1.5

Find the power delivered to an element at $t = 3\,\mathrm{ms}$ if the current entering its positive terminal is:

$$i(t) = 5\cos(60\pi t)\,\mathrm{A}$$

and the voltage is:

(a) $v = 3i$

(b) $v = 3\frac{di}{dt}$

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Power and Energy Examples

Practice Problem 1.5

Find the power delivered to the element in [Example 1.5] at

$$t = 5\,\text{ms}$$

if the current remains the same but the voltage is:

(a) $v = 6i\ \text{V}$

(b) $v = \left(6 + 10\int_0^t i\,dt\right)\ \text{V}$

Example 1.6

How much energy does a 100-W electric bulb consume in two hours?

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Power and Energy Examples

Practice Problem 1.6

A home electric heater draws 12 A when connected to a 115 V outlet. How much energy is consumed by the heater over a period of 24 hours?

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Circuit Elements Example

Example 1.7

Calculate the power supplied or absorbed by each element in Fig. 1.15.

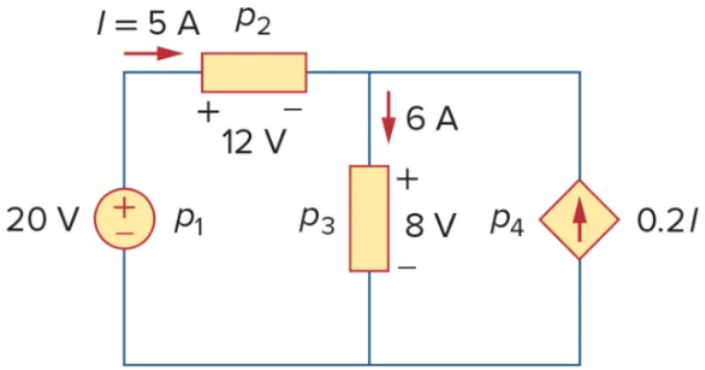

Figure 1.15
For Example 1.7.

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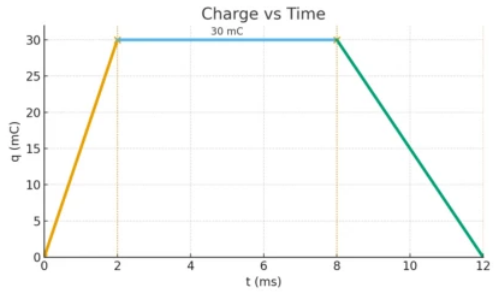
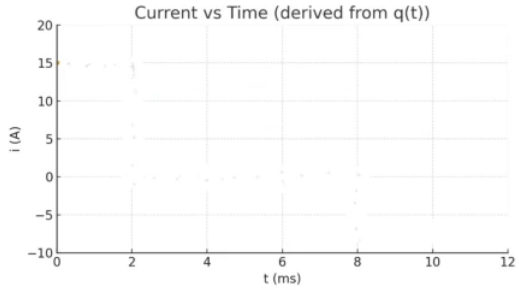
Extra Examples for charge and current

The charge entering a certain element is shown in the figure below.
Find the current at:
(a) t = 2 ms **(b)** t = 6 ms **(c)** t = 8 ms
(Assume q(t) is in mC and t is in ms.) and Sketch the corresponding current.





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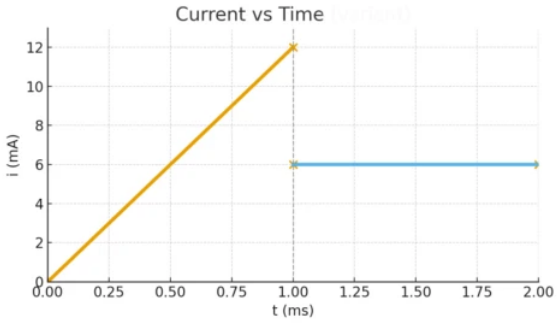
Example

The current flowing past a point in a device is shown in Fig. P1.8

- From 0 to $1\,\text{ms}$, $i(t)$ increases linearly from 0 to $12\,\text{mA}$.
- At $t = 1\,\text{ms}$ the current steps down.
- From 1 to $2\,\text{ms}$, $i(t) = 6\,\text{mA}$ (constant).

Calculate the total charge that passes the point over $0 \le t \le 2\,\text{ms}$.

(Use $q = \int i\,dt$; note $1\,\text{mA}\cdot\text{ms} = 1\,\mu\text{C}$)



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Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Problemset 1-Chapter 1- Basic Concepts

Name: _____ Marks: _____

Chapter 1, Problem 1.

How many coulombs are represented by these amounts of electrons?

(a) 6.482×10^{17} (b) 1.24×10^{18}

(c) 2.46×10^{19} (d) 1.628×10^{20}

Chapter 1, Problem 2.

Determine the current flowing through an element if the charge flow is given by

(a) $q(t) = (3t + 8)$ mC

(b) $q(t) = (8t^2 + 4t - 2)$ C

(c) $q(t) = (3e^{-t} - 5e^{-2t})$ nC

(d) $q(t) = 10\sin(120\pi t)$ pC

(e) $q(t) = 20e^{-4t}\cos(50t)$ μC

Chapter 1, Problem 3.

Find the charge $q(t)$ flowing through a device if the current is:

(a) $i(t) = 3\text{A}, q(0) = 1\text{C}$

(b) $i(t) = (2t + 5)\text{mA}, q(0) = 0$

(c) $i(t) = 20\cos(10t + \pi/6)\mu\text{A}, q(0) = 2\ \mu\text{C}$

Chapter 1, Problem 4.

A current of 7.4 A flows through a conductor. Calculate how much charge passes through any cross-section of the conductor in 20 seconds.

Chapter 1, Problem 5.

Determine the total charge transferred over the time interval of $0 \le t \le 10\text{s}$ when $i(t) = \frac{1}{2}t$ A.

Chapter 1, Problem 6.

The charge entering a certain element is shown in Fig. 1.23. Find the current at:

(a) $t = 1$ ms(b) $t = 6$ ms (c) $t = 10$ ms

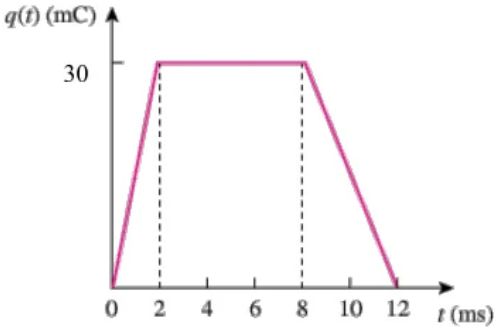


Figure 1.23
For Prob. 1.6.

Chapter 1, Problem 7.

The charge flowing in a wire is plotted in Fig. 1.24. Sketch the corresponding current.

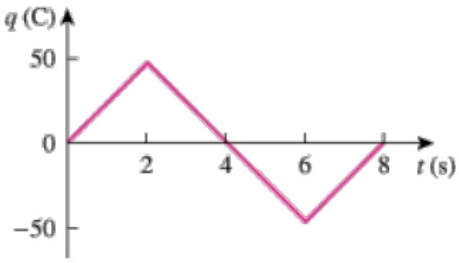


Figure 1.24
For Prob. 1.7.

Chapter 1, Problem 8.

The current flowing past a point in a device is shown in Fig. 1.25. Calculate the total charge through the point.

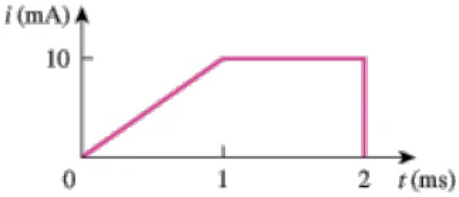


Figure 1.25
For Prob. 1.8.

Chapter 1, Problem 9.

The current through an element is shown in Fig. 1.26. Determine the total charge that passed through the element at:

(a) $t = 1$ s (b) $t = 3$ s (c) $t = 5$ s

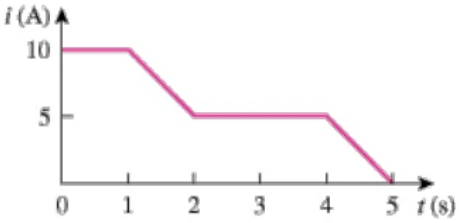


Figure 1.26
For Prob. 1.9.

Chapter 1, Problem 10.

A lightning bolt with 10 kA strikes an object for 15 μs. How much charge is deposited on the object?

Chapter 1, Problem 11.

A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h.
How much charge can it release at that rate? If its terminals voltage is 1.5 V, how much energy can the battery deliver?

Chapter 1, Problem 15.

The current entering the positive terminal of a device is $i(t) = 6e^{-2t}$ mA and the voltage across the device is $v(t) = 10\,di/dt$ V.

(a) Find the charge delivered to the device between $t = 0$ and $t = 2$ s.
(b) Calculate the power absorbed.
(c) Determine the energy absorbed between t = 0 and t = 3 s.

Chapter 1, Problem 17.

Figure 1.28 shows a circuit with five elements. If $p_1 = -205$ W, $p_2 = 60$ W, $p_4 = 45$ W, and $p_5 = 30$ W, calculate the power p_3 absorbed by element 3.

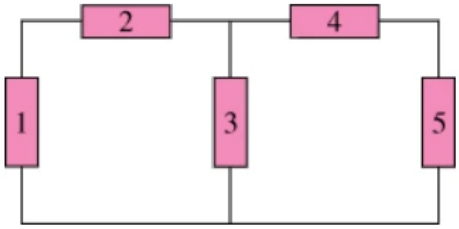


Figure 1.28
For Prob. 1.17.

Chapter 1, Problem 18.

Find the power absorbed by each of the elements in Fig. 1.29.

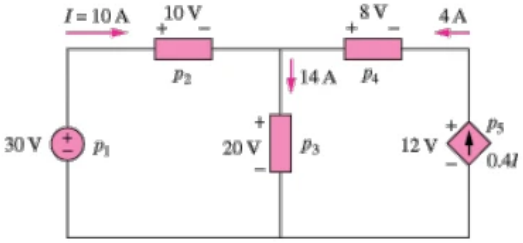


Figure 1.29
For Prob. 1.18

Chapter 1, Problem 19.

Find I and power absorbed by each element in the network of Fig. 1.30.

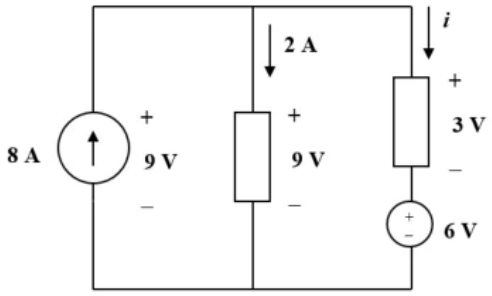


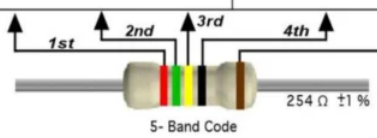
Figure 1.30
For Prob. 1.19.

Resistor Colour Code

- The resistor color code uses colors to represent numbers.
- Each color on the chart is equal to a number or percentage.
- Thus, these color bands provide a means to identify the resistor's resistance value and tolerance.

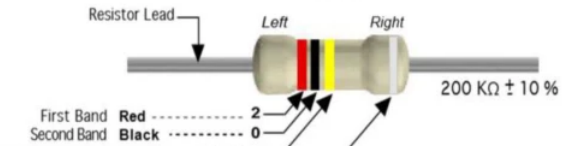
4- Band Code

Color	1st Band	2nd Band	3rd Band	Decimal Multiplier	Tolerance
Black	0	0	0	1	
Brown	1	1	1	10	± 1%
Red	2	2	2	100	± 2%
Orange	3	3	3	1K	1,000
Yellow	4	4	4	10K	10,000
Green	5	5	5	100K	100,000
Blue	6	6	6	1M	1,000,000
Violet	7	7	7	10M	10,000,000
Gray	8	8	8		100,000,000
White	9	9	9		1,000,000,000
Gold				0.1	± 5%
Silver				0.01	± 10%
None					± 20%



5- Band Code

Calculation



First Band **Red** ---------- **2**
Second Band **Black** ---------- **0**
Multiplier Band **Yellow** -- **x10,000**
Tolerance Band **Silver** -------**10** %

The Gold or Silver band is always placed to the right.
The resistor value is read from the left to right.

If there is no tolerance band, then find the side that has a band closest to a lead and make that the first band.

Equation

20 x 10,000 = 200,000

1,000 = 1K

Resistor = 200 K Ω
with a ± 10 % Tolerance

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Conductance

- What is conductance?
  - Measures how well a material will conduct current.
  - It is reciprocal of resistance, denoted with symbol, G
  - Measured in Siemens (S)

$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

In symbol notation:

$$G = \frac{1}{R} \quad \text{(measured in siemens, S)}$$

Ω^{-1}

- Example: 1M Ohm resistance, is 10^{-6}S conductance
- An Imp. Note:
  - Large the conductance, less the resistance
  - Thus greater the conductivity

$G = \frac{1}{1M} = 10^{-6}\,S$

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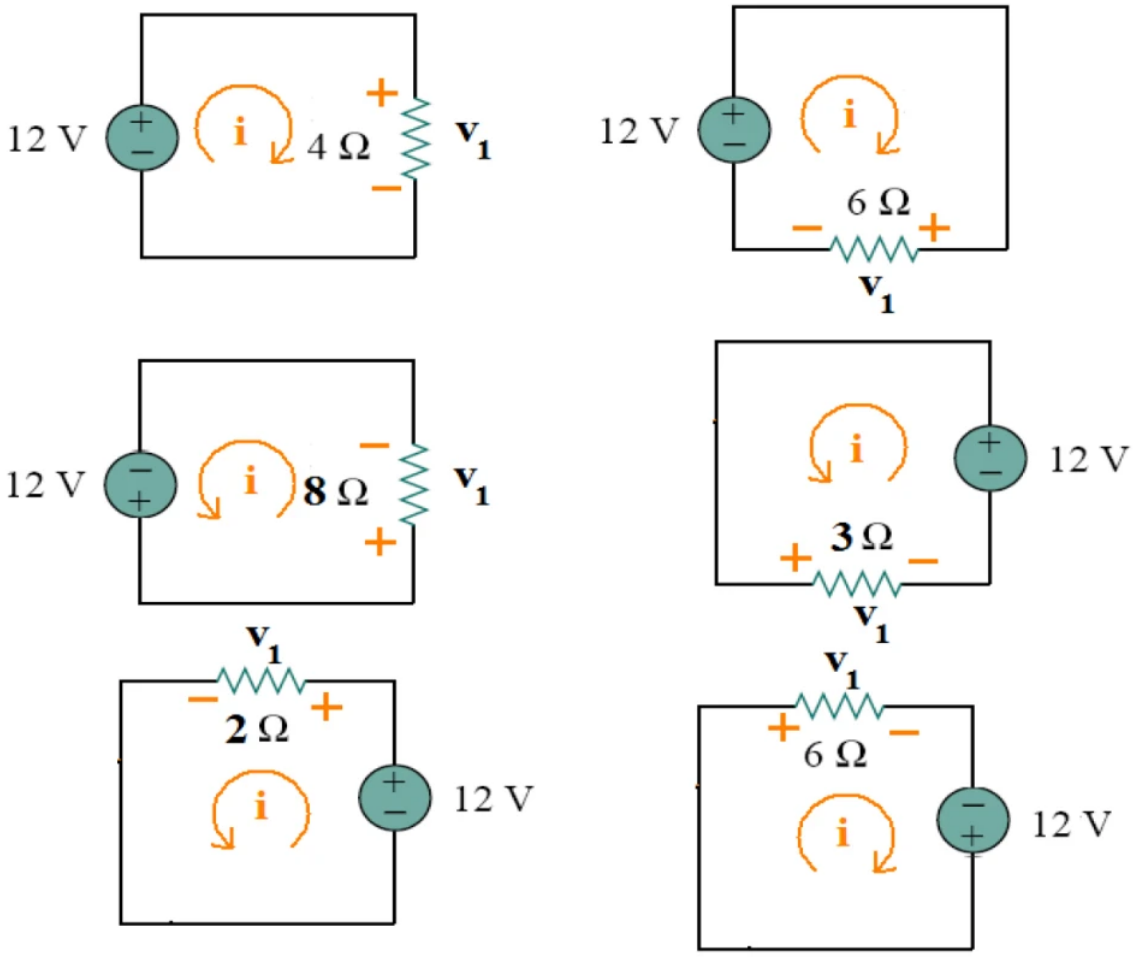

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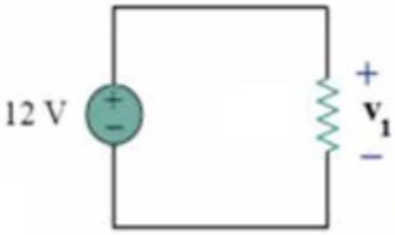
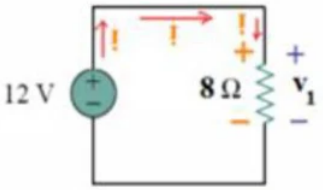
Voltage Sign

If a voltage sign in already marked on a resistor (circuit)
we must check whether the voltage sign conforms to the current direction or not.

Example 1

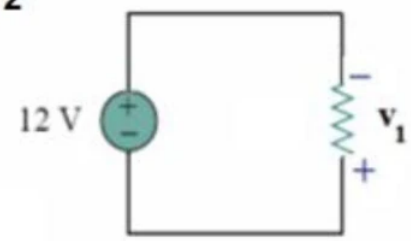


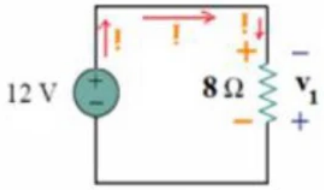


$v_1 = 12$ V

If the sign conforms to the current direction then it is OK.

Example 2





$v_1 = -12$ V

If the sign does not conform to the current direction,
the voltage value will be taken (written) as negative.

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Examples

Example 2.1

An electric iron operates at 120 V and draws a current of 2 A. Find its resistance.

I=V/R

Practice Problem 2.1

A toaster has a heating element with resistance of 15 Ω. If it is connected to a 110 V supply, how much current does it draw?

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Examples

Example 2.2

In the circuit shown in Fig. 2.8, calculate the current i, the conductance G, and the power p.

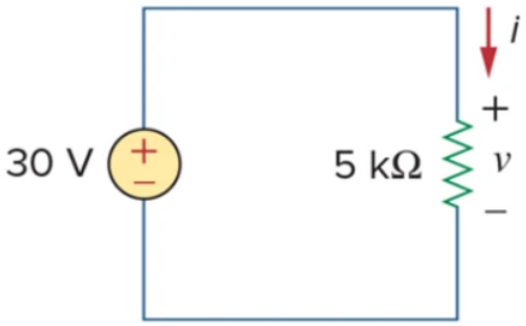

Figure 2.8
For Example 2.2.

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Examples

Practice Problem 2.2

For the circuit shown in Fig. 2.9, calculate the voltage v, the conductance G, and the power p.

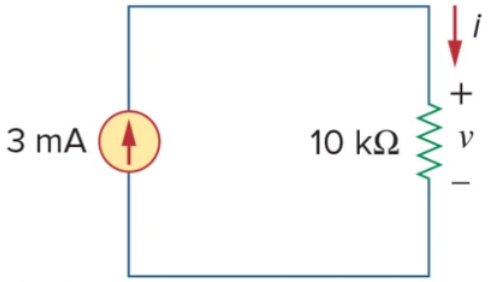

Figure 2.9
For Practice Prob. 2.2

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Examples

Example 2.3

A voltage source of 20sinπt V is connected across a 5-kΩ resistor. Find the current through the resistor and the power dissipated.

Practice Problem 2.3

A resistor absorbs an instantaneous power of $30\cos^2 t$ mW when connected to a voltage source v=15cost V. Find i and R.

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Examples

Example 2.4

Determine the number of branches and nodes in the circuit shown in Fig. 2.12. Identify which elements are in series and which are in parallel.

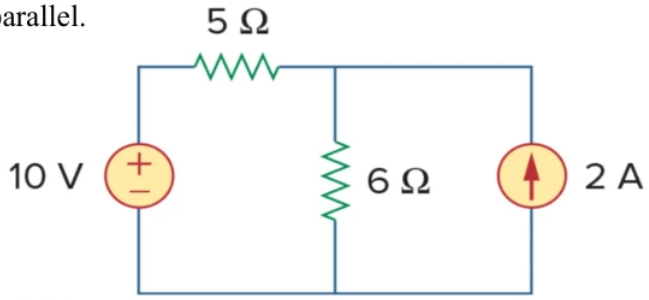

Figure 2.12
For Example 2.4.

Solution: Since there are four elements in the circuit, the circuit has four branches: 10 V, 5 Ω, 6 Ω, and 2 A. The circuit has three nodes as identified in Fig. 2.13. The 5-Ω resistor is in series with the 10-V voltage source because the same current would flow in both. The 6-Ω resistor is in parallel with the 2-A current source because both are connected to the same nodes 2 and 3.

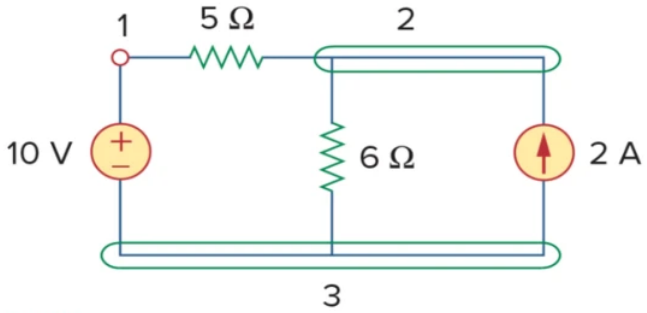

Figure 2.13

The three nodes in the circuit of Fig. 2.12.

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Examples

Practice Problem 2.4

How many branches and nodes does the circuit in Fig. 2.14 have? Identify the elements that are in series and in parallel.

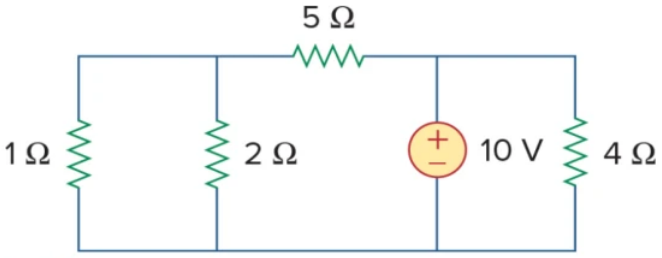

Figure 2.14
For Practice Prob. 2.4.

Answer: Five branches and three nodes are identified in Fig. 2.15. The 1-Ω and 2-Ω resistors are in parallel. The 4-Ω resistor and 10-V source are also in parallel.

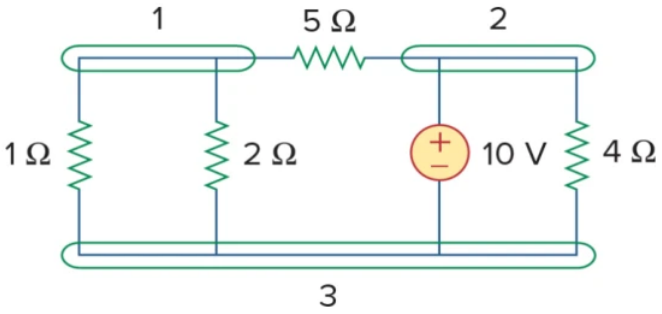

Figure 2.15
Answer for Practice Prob. 2.4.

5 branches and **3** nodes. The 1 ohm and 2 ohm resistors are in parallel. The 4 ohm resistor and the 10 volt source are also in parallel.

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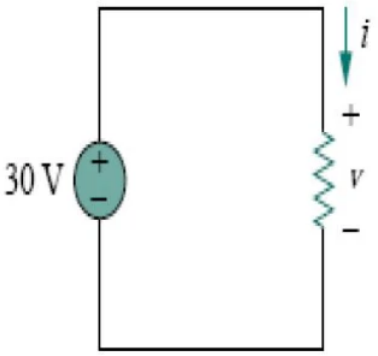
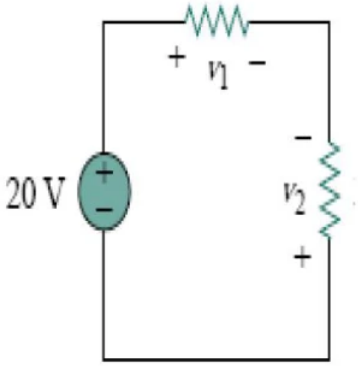
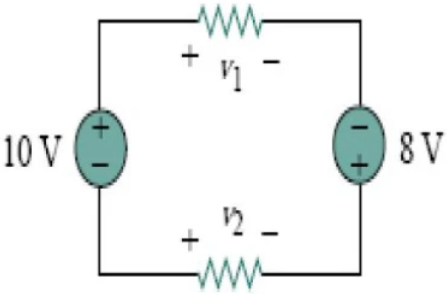
Let us do some practice of writting voltage equations.



-30 + v = 0



- 20 + v1 - v2 = 0



- 10 + v1 - 8 - v2 = 0

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Examples

Example 2.5

For the circuit in Fig. 2.21(a), find voltages V1and V2

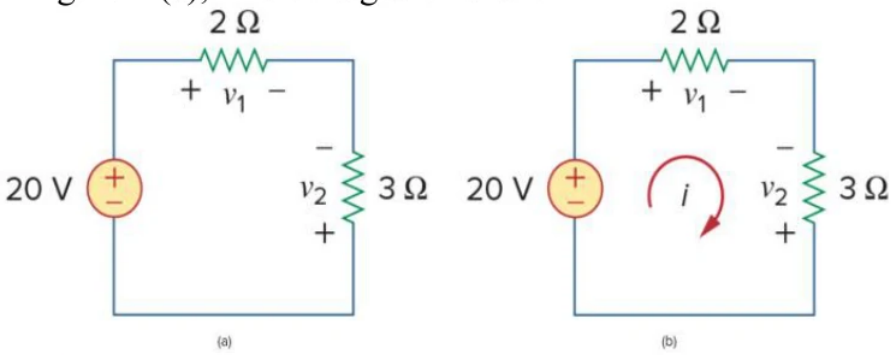

Figure 2.21
For Example 2.5.

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Practice Problem 2.5

Find v_1 and v_2 in the circuit of Fig. 2.22.

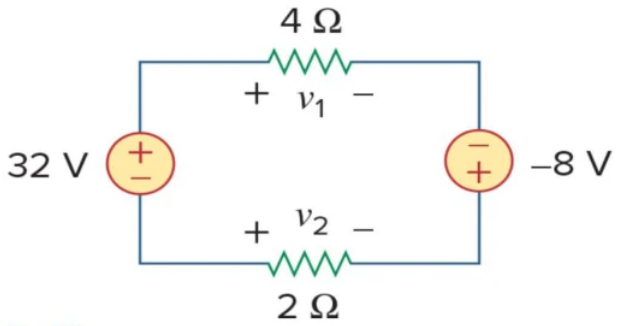

Figure 2.22
For Practice Prob. 2.5.

Answer: 16 V, –8 V.

Applying **KVL** to the loop we get:

$$-32 + 4i - (-8) + 2i = 0 \text{ which leads to } i = 24/6 = 4A$$

$$v_1 = 4i = \mathbf{16\ V} \quad \text{and} \quad v_2 = -2i = \mathbf{-8\ V}$$

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Examples

Example 2.6

Determine v_o and i in the circuit shown in Fig. 2.23(a).

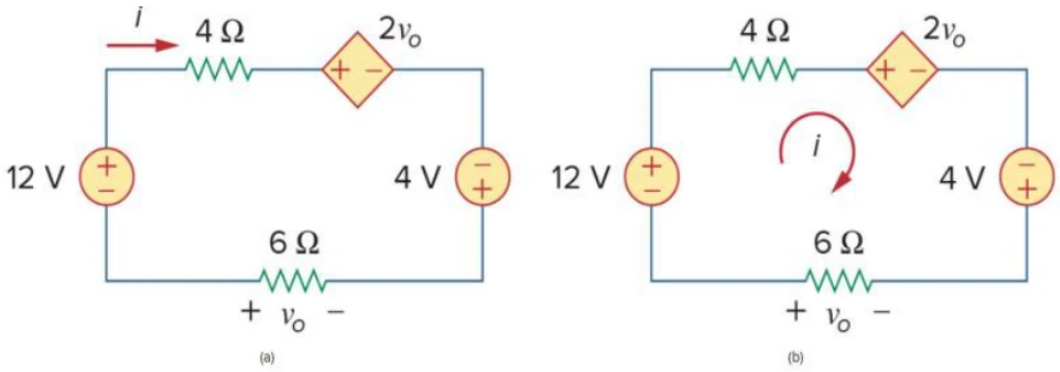

Figure 2.23
For Example 2.6.

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Examples

Example 2.7

Find current i_o and voltage v_o in the circuit shown in Fig. 2.25.

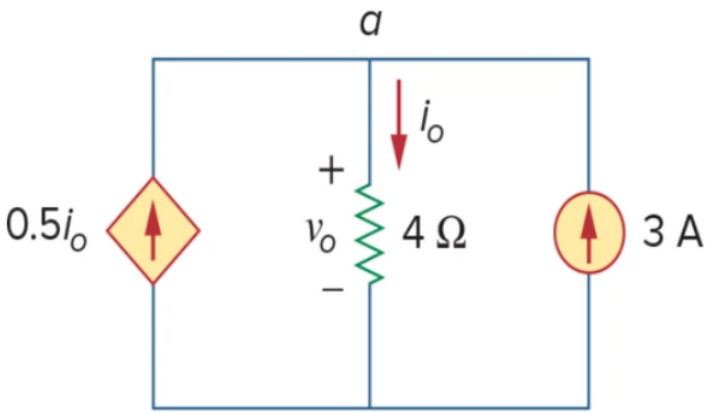

Figure 2.25
For Example 2.7.

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Practice Problem 2.6

Find v_x and v_o in the circuit of Fig. 2.24.

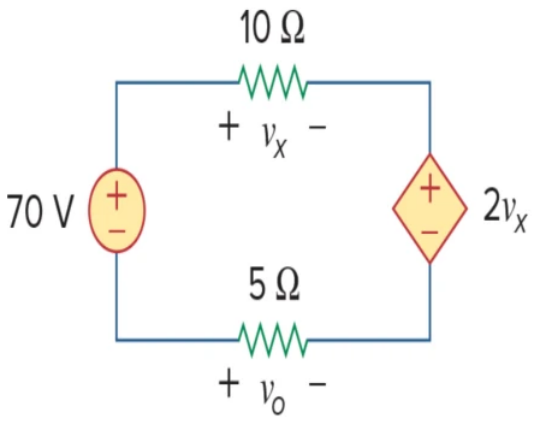

Figure 2.24
For Practice Prob. 2.6.

Answer: 20 V, –10 V.

Applying **KVL** to the loop we get:

$$-70 + 10i + 2v_x + 5i = 0$$

But, $v_x = 10i$ and $v_0 = -5i$. Hence,

$-70 + 10i + 20i + 5i = 0$ which leads to $i = 2$ A.

Thus, v_x = **20V** and v_0 = **–10 V**

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Examples

Practice Problem 2.7

Find v_o and i_o in the circuit of Fig. 2.26.

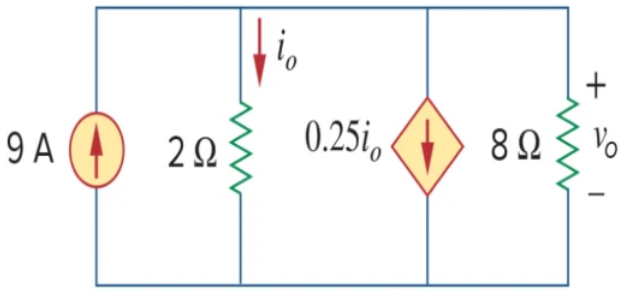

Figure 2.26
For Practice Prob. 2.7.

Answer: 12 V, 6 A.

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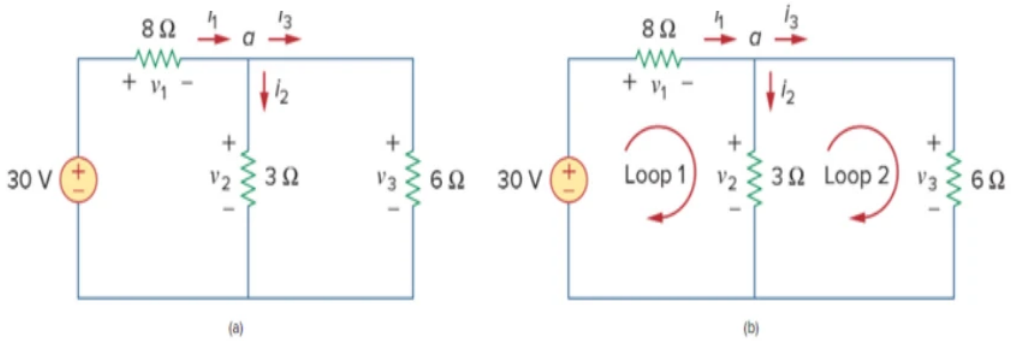
Examples

Example 2.8

Find currents and voltages in the circuit shown in Fig. 2.27(a).



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Example 2.9

Find R_{eq} for the circuit shown in Fig. 2.34.

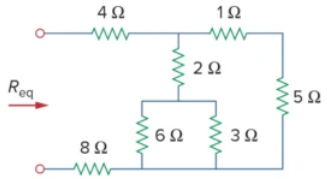
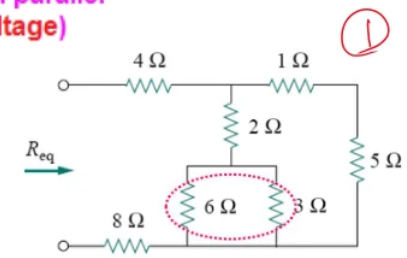


Figure 2.34
For Example 2.9.

Hint:

1. Solve for the parallel components first.
2. Solve starting from Right .

Identify the branches that are in parallel (same voltage)



Solution:

To get R_{eq}, we combine resistors in series and in parallel. The 6-Ω and 3-Ω resistors are in parallel, so their equivalent resistance is

$$6\ \Omega \parallel 3\Omega = \frac{6 \times 3}{6+3} = 2\ \Omega$$

(The symbol ∥ is used to indicate a parallel combination.) Also, the 1-Ω and 5-Ω resistors are in series; hence their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega$$

Thus the circuit in Fig. 2.34 is reduced to that in Fig. 2.35(a). In Fig. 2.35(a), we notice that the two 2-Ω resistors are in series, so the equivalent resistance is

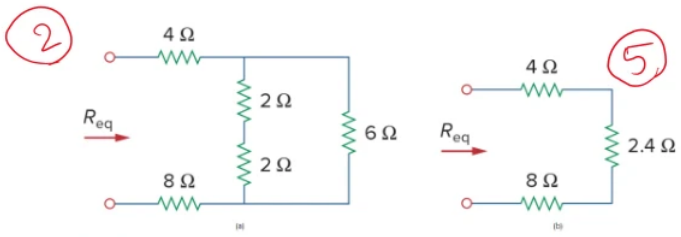
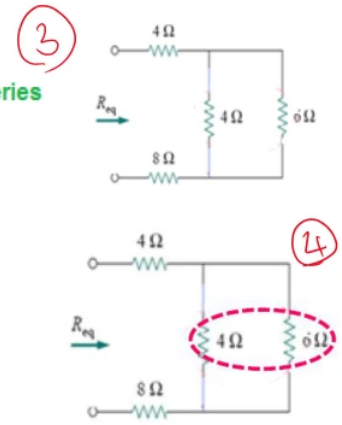


Figure 2.35
Equivalent circuits for Example 2.9.

Adding the series components



This 4-Ω resistor is now in parallel with the 6-Ω resistor in Fig. 2.35(a); their equivalent resistance is

$$4\ \Omega \parallel 6\ \Omega = \frac{4 \times 6}{4+6} = 2.4\ \Omega$$

The circuit in Fig. 2.35(a) is now replaced with that in Fig. 2.35(b). In Fig. 2.35(b), the three resistors are in series. Hence, the equivalent resistance for the circuit is

$$R_{eq} = 4\ \Omega + 2.4\ \Omega + 8\ \Omega = 14.4\ \Omega$$

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Example 2.10

Calculate the equivalent resistance R_{ab} in the circuit in Fig. 2.37.

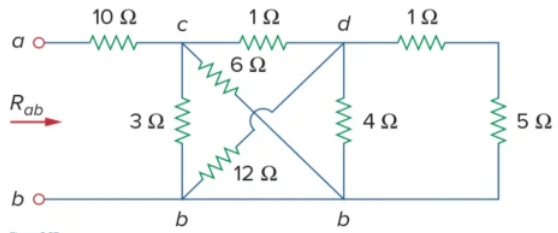


Figure 2.37
For Example 2.10.

Solution:

The 3-Ω and 6-Ω resistors are in parallel because they are connected to the same two nodes c and b. Their combined resistance is

$$3\ \Omega \parallel 6\ \Omega = \frac{3 \times 6}{3+6} = 2\ \Omega \tag{2.10.1}$$

Similarly, the 12-Ω and 4-Ω resistors are in parallel since they are connected to the same two nodes d and b. Hence

$$12\ \Omega \parallel 4\ \Omega = \frac{12 \times 4}{12+4} = 3\ \Omega \tag{2.10.2}$$

Also the 1-Ω and 5-Ω resistors are in series; hence, their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega \tag{2.10.3}$$

With these three combinations, we can replace the circuit in Fig. 2.37 with that in Fig. 2.38(a). In Fig. 2.38(a), 3-Ω in parallel with 6-Ω gives 2-Ω, as calculated in Eq. (2.10.1). This 2-Ω equivalent resistance is now in series with the 1-Ω resistance to give a combined resistance of $1\ \Omega + 2\ \Omega = 3\ \Omega$. Thus, we replace the circuit in Fig. 2.38(a) with that in Fig. 2.38(b). In Fig. 2.38(b), we combine the 2-Ω and 3-Ω resistors in parallel to get

$$2\ \Omega \parallel 3\ \Omega = \frac{2 \times 3}{2+3} = 1.2\ \Omega$$

This 1.2-Ω resistor is in series with the 10-Ω resistor, so that

$$R_{ab} = 10 + 1.2 = 11.2\ \Omega$$

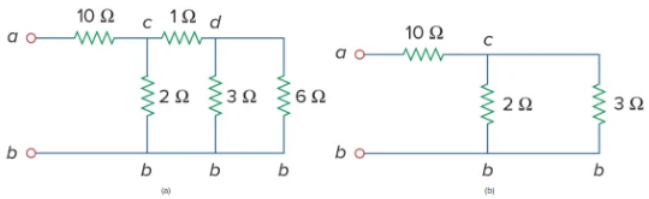
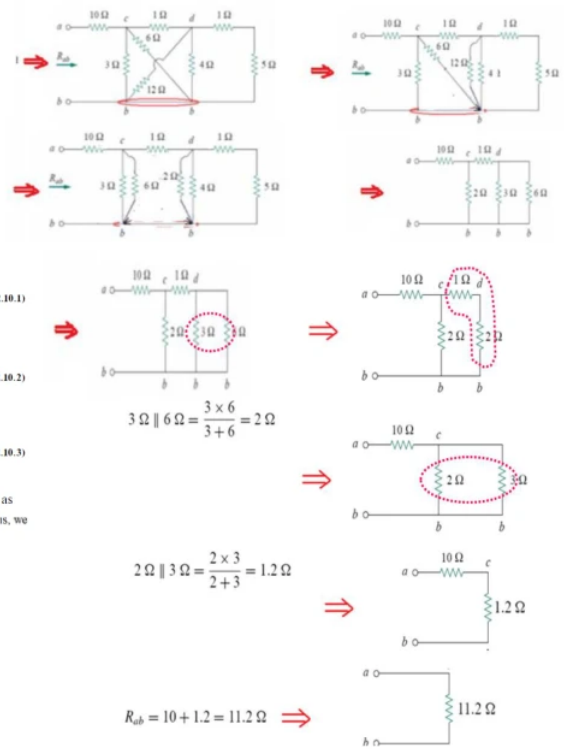


Figure 2.38
Equivalent circuits for Example 2.10.

1. Solve for the parallel components first.
2. Solve starting from Right .



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Practice Problem 2.10

Find R_{ab} for the circuit in Fig. 2.39.

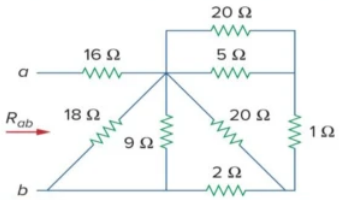
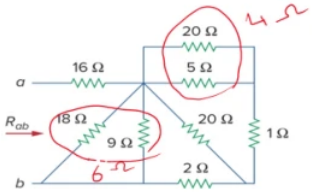


Figure 2.39
For Practice Prob. 2.10.



Answer: 19 Ω.

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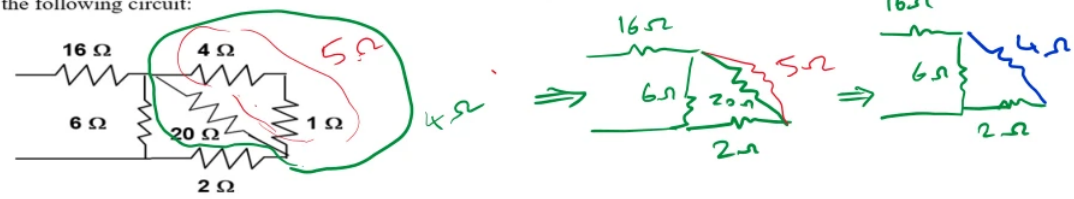
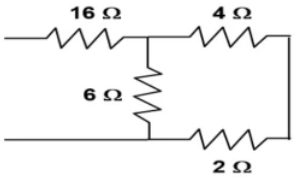
Combining the 9-ohm resistor and the 18-ohm resistor yields [9x18]/[9+18] = 6 ohms.

Combining the 5 ohm and the 20 ohm resistors in parallel produces [5x20/(5+20)] = 4 ohms We now have the following circuit:



The 4 ohm and 1 ohm resistors can be combined into a 5 ohm resistor in parallel with a 20 ohm resistor. This will result in [5x20/(5+20)] = 4 ohms and the circuit shown below:



The 4 ohm and 2 ohm resistors are in series and can be replaced by a 6 ohm resistor. This gives a 6 ohm resistor in parallel with a 6 ohm resistor, [6x6/(6+6)] = 3 ohms. We now have a 3 ohm resistor in series with a 16 ohm resistor or 3 + 16 = 19 ohms. Therefore:

$$R_{eq} = 19 \text{ ohms}$$

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Example 2.11

Find the equivalent conductance G_{eq} for the circuit in Fig. 2.40(a).

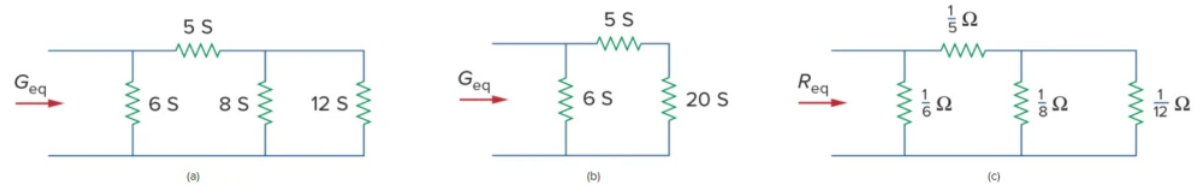

Figure 2.40
For Example 2.11: (a) original circuit, (b) its equivalent circuit, (c) same circuit as in (a) but resistors are expressed in ohms.

Solution:

The 8-S and 12-S resistors are in parallel, so their conductance is

$$8\text{ S} + 12\text{ S} = 20\text{ S}$$

This 20-S resistor is now in series with 5 S as shown in Fig. 2.40(b) so that the combined conductance is

$$\frac{20 \times 5}{20 + 5} = 4\text{ S}$$

This is in parallel with the 6-S resistor. Hence,

$$G_{eq} = 6 + 4 = 10\text{ S}$$

We should note that the circuit in Fig. 2.40(a) is the same as that in Fig. 2.40(c). While the resistors in Fig. 2.40(a) are expressed in siemens, those in Fig. 2.40(c) are expressed in ohms. To show that the circuits are the same, we find R_{eq} for the circuit in Fig. 2.40(c).

$$R_{eq} = \frac{1}{6} \Big\| \left(\frac{1}{5} + \frac{1}{8} \Big\| \frac{1}{12} \right) = \frac{1}{6} \Big\| \left(\frac{1}{5} + \frac{1}{20} \right) = \frac{1}{6} \Big\| \frac{1}{4}$$

$$G_{eq} = \frac{1}{R_{eq}} = 10\text{ S}$$

This is the same as we obtained previously.

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Example 2.12

Find i_o and v_o in the circuit shown in Fig. 2.42(a). Calculate the power dissipated in the 3-Ω resistor.

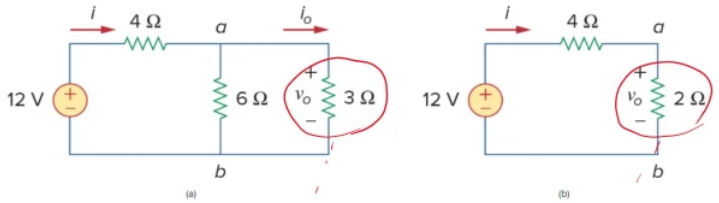


Figure 2.42
For Example 2.12: (a) original circuit, (b) its equivalent circuit.

Solution:

The 6-Ω and 3-Ω resistors are in parallel, so their combined resistance is

$$6\ \Omega \parallel 3\ \Omega = \frac{6\times 3}{6+3} = 2\ \Omega$$

Thus, our circuit reduces to that shown in Fig. 2.42(b). Notice that v_o is not affected by the combination of the resistors because the resistors are in parallel and therefore have the same voltage v_o. From Fig. 2.42(b), we can obtain v_o in two ways. One way is to apply Ohm's law to get

Page 50

$$i = \frac{12}{4+2} = 2\ \text{A}$$

and hence, $v_o = 2i = 2 \times 2 = 4$ V. Another way is to apply voltage division, since the 12 V in Fig. 2.42(b) is divided between the 4-Ω and 2-Ω resistors. Hence,

$$v_o = \frac{2}{2+4}(12\ \text{V}) = 4\ \text{V}$$

Similarly, i_o can be obtained in two ways. One approach is to apply Ohm's law to the 3-Ω resistor in Fig. 2.42(a) now that we know v_o; thus,

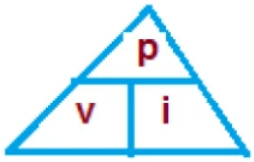
$$v_o = 3i_o = 4 \quad \Rightarrow \quad i_o = \frac{4}{3}\ \text{A}$$

Another approach is to apply current division to the circuit in Fig. 2.42(a) now that we know i, by writing

$$i_o = \frac{6}{6+3}i = \frac{2}{3}(2\ \text{A}) = \frac{4}{3}\ \text{A}$$

The power dissipated in the 3-Ω resistor is

$$p_o = v_o i_o = 4\left(\frac{4}{3}\right) = 5.333\ \text{W}$$



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Example 2.13

For the circuit shown in Fig. 2.44(a), determine: (a) the voltage v_o, (b) the power supplied by the current source, (c) the power absorbed by each resistor.

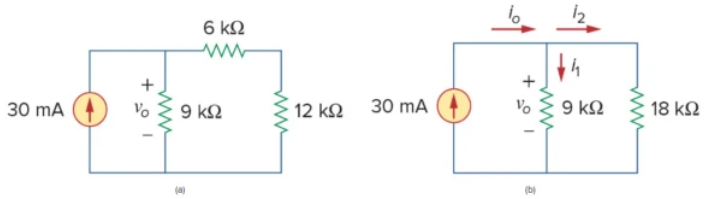

Figure 2.44
For Example 2.13: (a) original circuit, (b) its equivalent circuit.

Solution:

(a) The 6-kΩ and 12-kΩ resistors are in series so that their combined value is $6 + 12 = 18$ kΩ. Thus the circuit in Fig. 2.44(a) reduces to that shown in Fig. 2.44(b). We now apply the current division technique to find i_1 and i_2.

$$i_1 = \frac{18{,}000}{9{,}000 + 18{,}000}(30\text{ mA}) = 20\text{ mA}$$
$$i_2 = \frac{9{,}000}{9{,}000 + 18{,}000}(30\text{ mA}) = 10\text{ mA}$$

Notice that the voltage across the 9-kΩ and 18-kΩ resistors is the same, and $v_o = 9{,}000 i_1 = 18{,}000 i_2 = 180$ V, as expected.

Page 51

(b) Power supplied by the source is

$$p_o = v_o i_o = 180\,(30)\text{ mW} = 5.4\text{ W}$$

(c) Power absorbed by the 12-kΩ resistor is

$$p = iv = i_2\,(i_2 R) = i_2^2 R = \left(10 \times 10^{-3}\right)^2 (12{,}000) = 1.2\text{ W}$$

Power absorbed by the 6-kΩ resistor is

$$p = i_2^2 R = \left(10 \times 10^{-3}\right)^2 (6{,}000) = 0.6\text{ W}$$

Power absorbed by the 9-kΩ resistor is

$$p = \frac{v_o^2}{R} = \frac{(180)^2}{9{,}000} = 3.6\text{ W}$$

or

$$p = v_o i_1 = 180\,(20)\text{ mW} = 3.6\text{ W}$$

Notice that the power supplied (5.4 W) equals the power absorbed ($1.2 + 0.6 + 3.6 = 5.4$ W). This is one way of checking results.

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Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Problemset 2-Chapter 2- Basic Laws

Name: _____ Marks: _____

Chapter 2, Problem 2.

Find the hot resistance of a light bulb rated 60W, 120 V.

[Hint: $p = v^2/R \rightarrow \quad \mathbf{R} = v^2/p$]

Chapter 2, Problem 4.

(a) Calculate current *i* in Fig. 2.68 when the switch is in position 1.

(b) Find the current when the switch is in position 2.

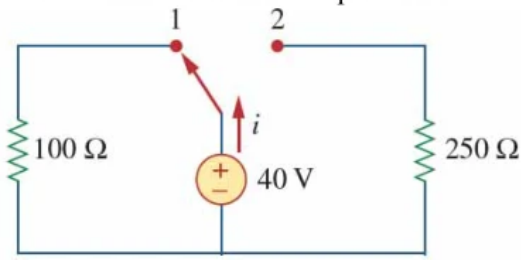

Figure 2.68
For Prob. 2.4.

Chapter 2, Problem 9.

Find i_1, i_2, and i_3 in Fig. 2.73.

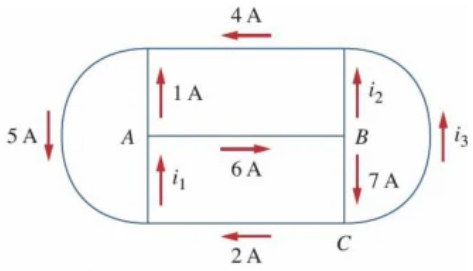

Figure 2.73
For Prob. 2.9.

Chapter 2, Problem 10.

Determine i_1 and i_2 in the circuit of Fig. 2.74.

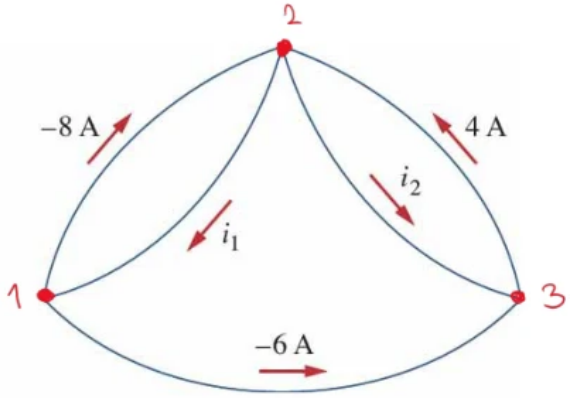

Figure 2.74
For Prob. 2.10.

Chapter 2, Problem 11.

In the circuit of Fig. 2.75, calculate V_1 and V_2.

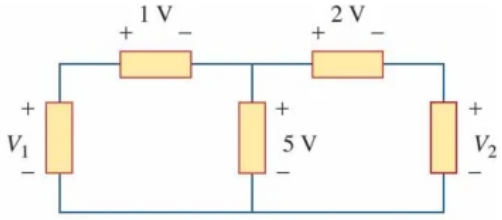

Figure 2.75
For Prob. 2.11.

Chapter 2, Problem 13.

For the circuit in Fig. 2.77, use KCL to find the branch currents I_1 to I_4.

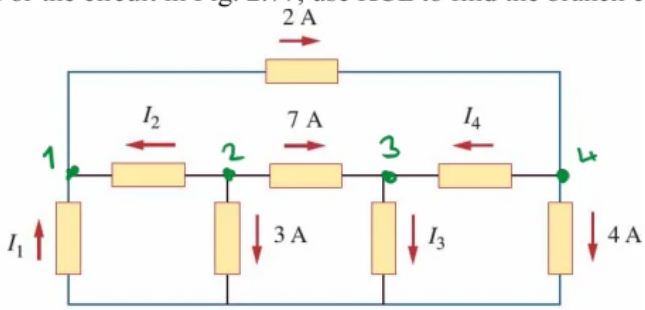

Figure 2.77
For Prob. 2.13.

Chapter 2, Problem 14.

Given the circuit in Fig. 2.78, use KVL to find the branch voltages V_1 to V_4.

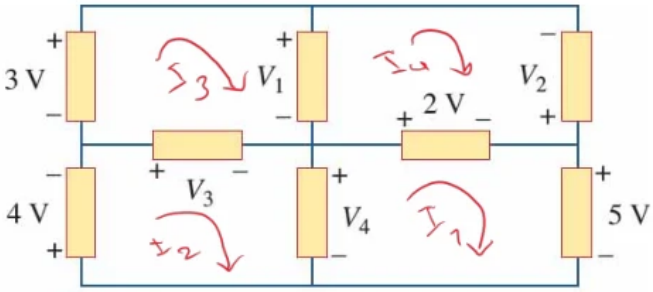

Figure 2.78
For Prob. 2.14.

Chapter 2, Problem 15.

Calculate v and i_x in the circuit of Fig. 2.79.

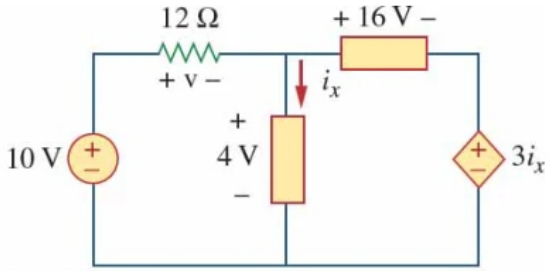

Figure 2.79
For Prob. 2.15.

Chapter 2, Problem 16.

Determine V_o in the circuit in Fig. 2.80.

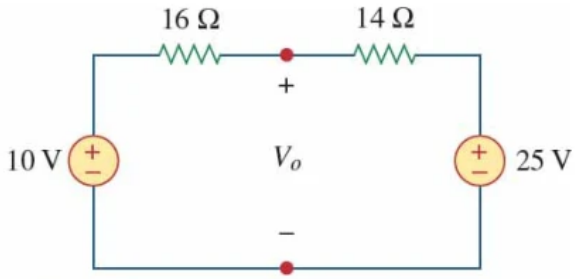

Figure 2.80
For Prob. 2.16.

Chapter 2, Problem 18.

Find I and V_{ab} in the circuit of Fig. 2.82.

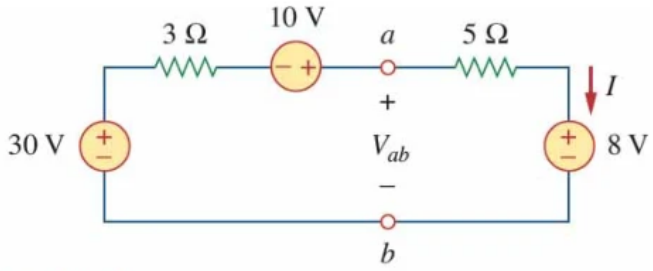

Figure 2.82
For Prob. 2.18.

Chapter 2, Problem 19.

From the circuit in Fig. 2.83, find I, the power dissipated by the resistor, and the power supplied by each source.

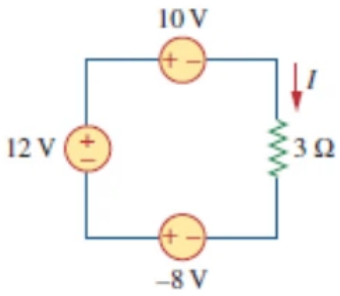

Figure 2.83
For Prob. 2.19.

Apply Nodal Analysis

Example 3.1

Calculate the node voltages in the circuit shown in Fig. 3.3(a).

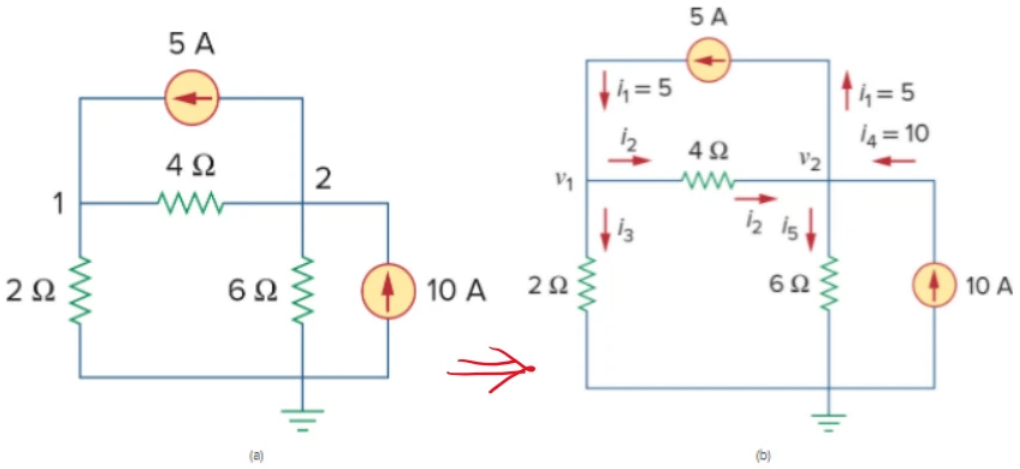

Figure 3.3
For Example 3.1: (a) original circuit, (b) circuit for analysis.

Solution:

Consider Fig. 3.3(b), where the circuit in Fig. 3.3(a) has been prepared for nodal analysis. Notice how the currents are selected for the application of KCL. Except for the branches with current sources, the labeling of the currents is arbitrary but consistent. (By consistent, we mean that if, for example, we assume that i_2 enters the 4-Ω resistor from the left-hand side, i_2 must leave the resistor from the right-hand side.) The reference node is selected, and the node voltages v_1 and v_2 are now to be determined.

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Practice Problem 3.1

Obtain the node voltages in the circuit of Fig. 3.4.

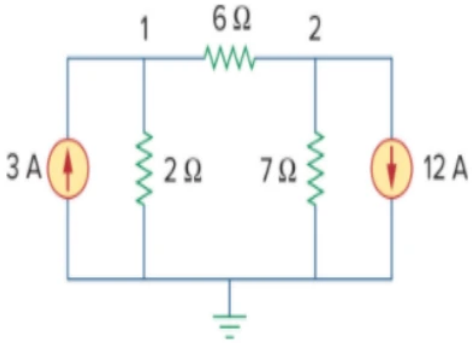

Figure 3.4
For Practice Prob. 3.1.

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Example 3.2

Determine the voltages at the nodes in Fig. 3.5(a).

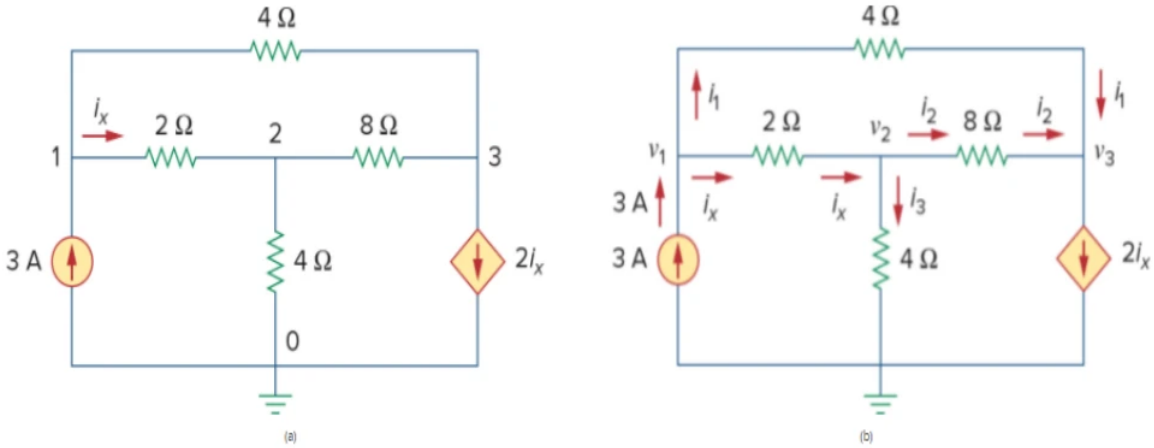

Figure 3.5
For Example 3.2: (a) original circuit, (b) circuit for analysis.

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Practice Problem 3.2

Find the voltages at the three nonreference nodes in the circuit of Fig. 3.6.

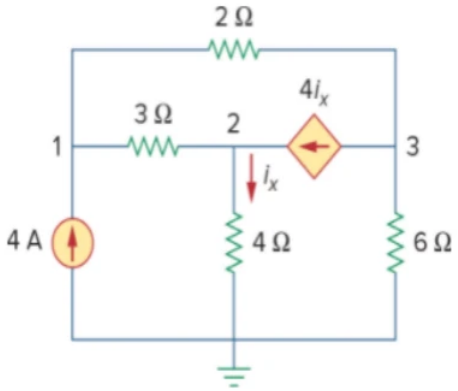
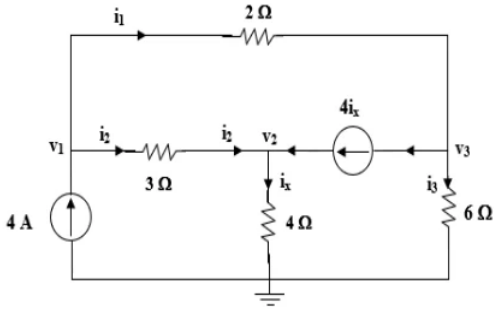


Figure 3.6
For Practice Prob. 3.2.



© McGraw Hill

Example 3.5

For the circuit in Fig. 3.18, find the branch currents I_1, I_2, and I_3 using mesh analysis.

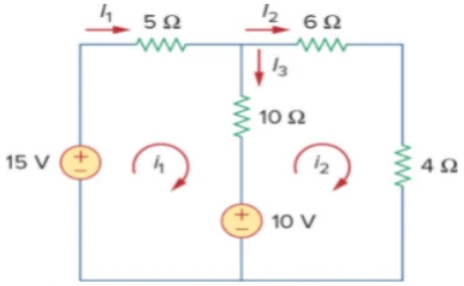


Figure 3.18
For Example 3.5.

© McGraw Hill

Example 3.6

Use mesh analysis to find the current I_o in the circuit of Fig. 3.20.

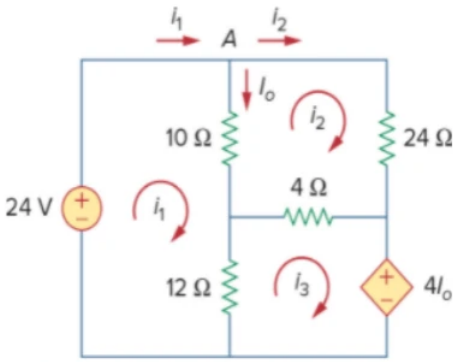


Figure 3.20
For Example 3.6.

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Practice Problem 3.6

Using mesh analysis, find I_o in the circuit of Fig. 3.21.

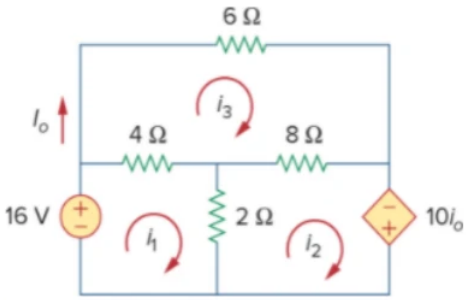

Figure 3.21
For Practice Prob. 3.6.

Answer: −4 A.

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ECET 250E-Linear Circuit 1
Problemset 3-Chapter 3- Methods of Analysis

Name: _____ Marks: _____

---

1. Chapter 3, Problem 1.

Given $\mathbf{R_1 = 4\ k\Omega}$, $\mathbf{R_2 = 2\ k\Omega}$, and $\mathbf{R_3 = 2\ k\Omega}$, determine the value of $\mathbf{I_x}$ using nodal analysis.

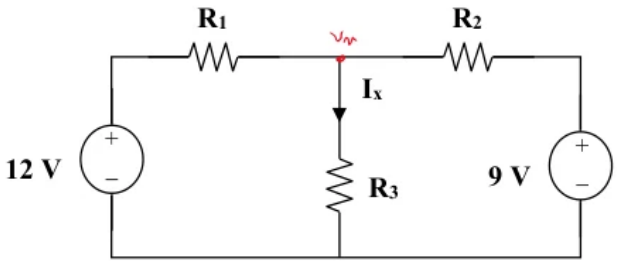


Figure 3.50
For Prob. 3.1

2. Chapter 3, Problem 2.

For the circuit in Fig. 3.51, obtain v_1 and v_2 using nodal analysis.

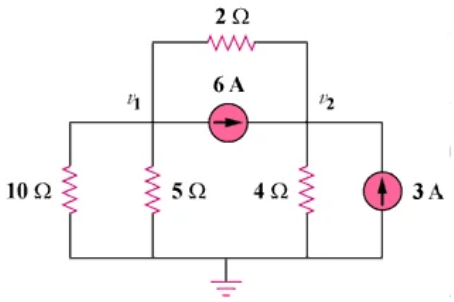


Figure 3.51
For Prob. 3.2.

3. Chapter 3, Problem 3.

Find the currents i_1 through i_4 and the voltage v_o in the circuit in Fig. 3.52 using nodal analysis.

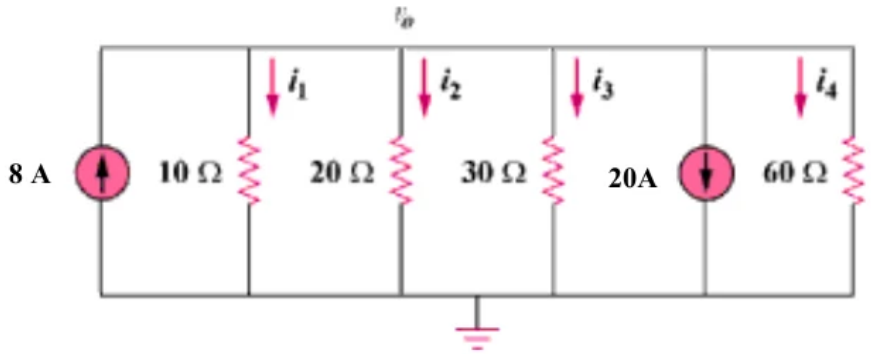


Figure 3.52
For Prob. 3.3.

4. Chapter 3, Problem 4

Given the circuit in Fig. 3.53, calculate the currents i_1 through i_4 using nodal analysis.

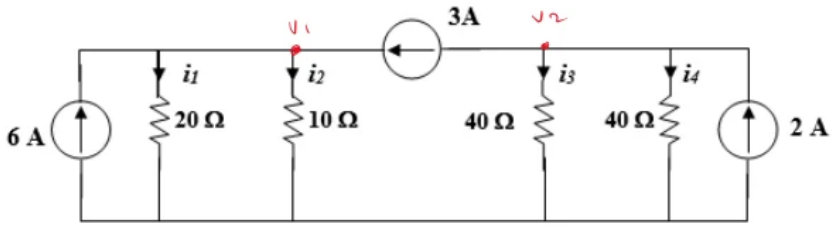


Figure 3.53
For Prob. 3.4.

5. Chapter 3, Problem 5.

Obtain v_0 in the circuit of Fig. 3.54 using Mesh analysis.

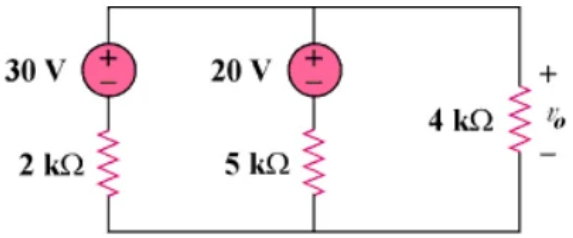


Figure 3.54
For Prob. 3.5.

6. Chapter 3, Problem 6.

Solve for V_1, in Fig. 3.55, using Mesh analysis.

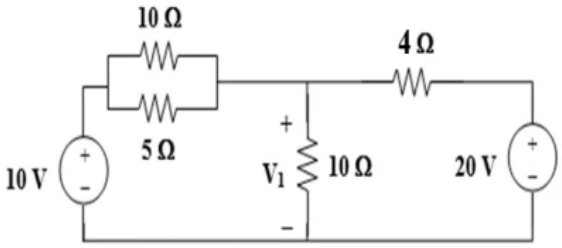


Figure 3.55
For Prob. 3.6.

7. Chapter 3, Problem 37.

Using mesh analysis, find v_0 in the circuit in Fig. 3.57.

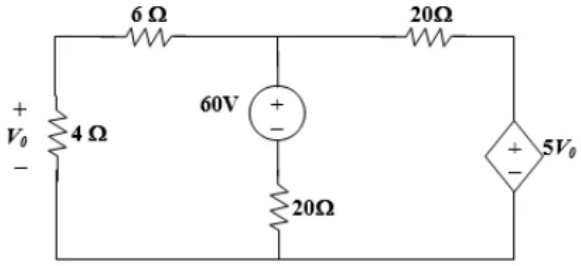


Figure
For Prob. 3.37.

8. Chapter 3, Problem 41.

Apply mesh analysis to find ***i*** in Fig. 3.87.

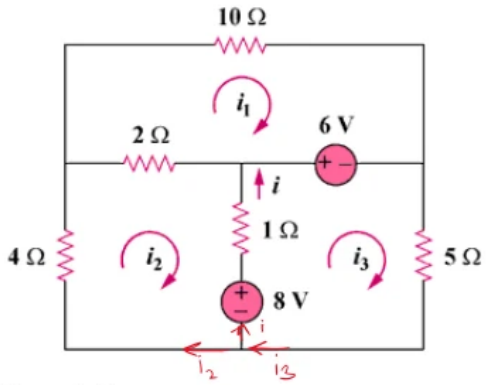


Figure 3.87
For Prob. 3.41.

9. Chapter 3, Problem 54

Given the circuit Fig. A, use mesh analysis to find the mesh currents.

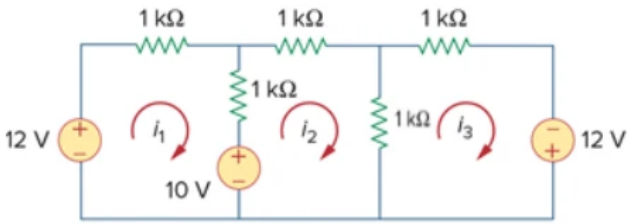
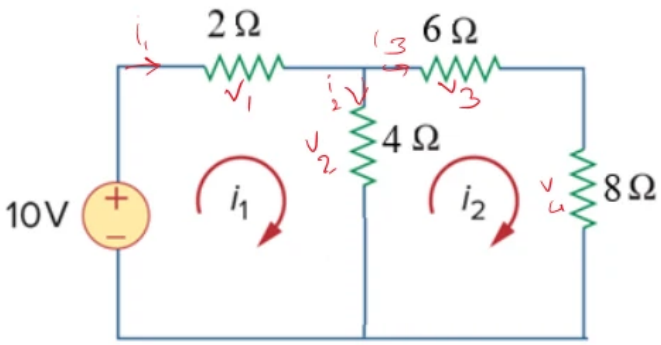


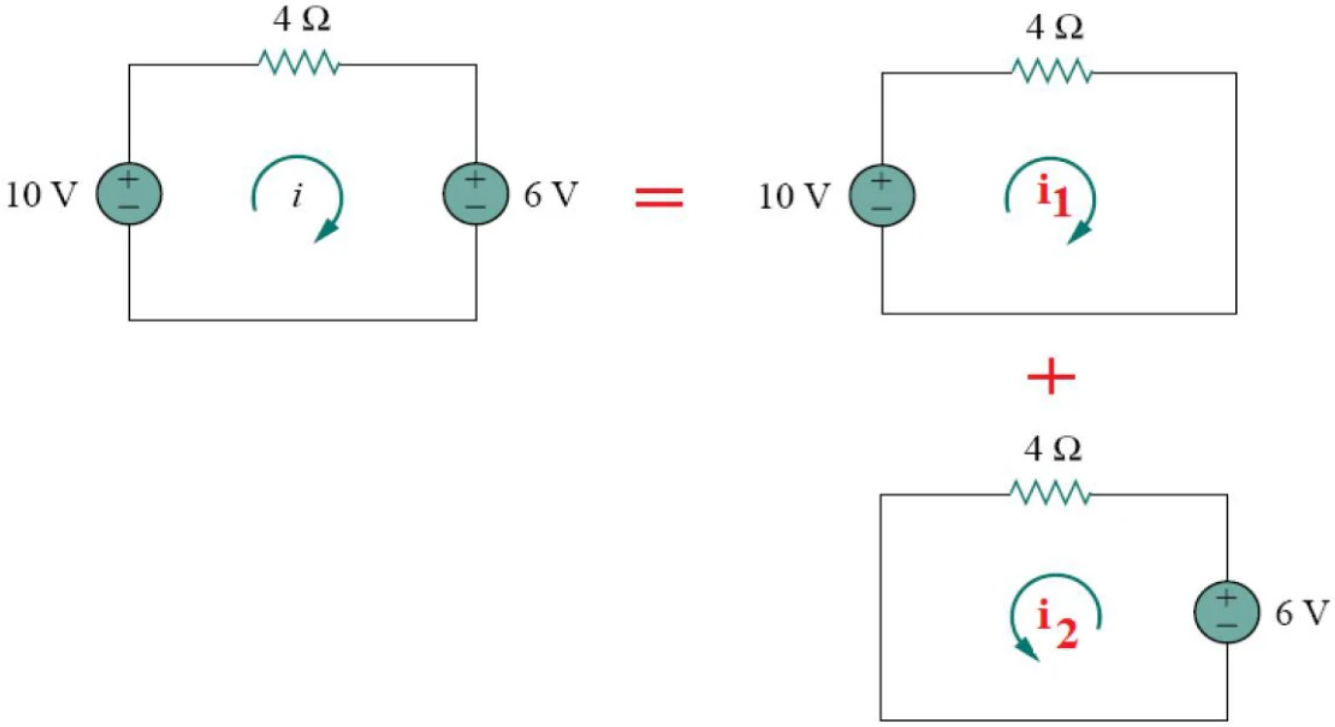
Figure 3.99
For Prob. 3.54

10. Chapter 3, Problem x

Given the circuit below, use mesh analysis to find the mesh currents voltages v_1, v_2 and v_3.



If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use nodal or mesh analysis as in Chapter 3. Another way is to determine the contribution of each independent source to the variable and then add them up. The latter approach is known as the *superposition*.



© McGraw Hill

Example 4.3

Use the superposition theorem to find v in the circuit of Fig. 4.6.

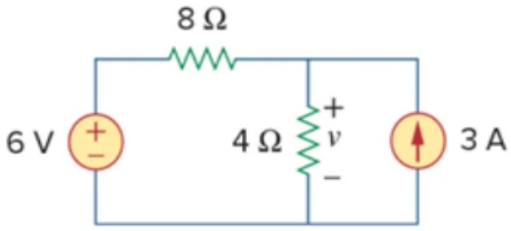


Figure 4.6
For Example 4.3.

© McGraw Hill

Practice Problem 4.3

Using the superposition theorem, find v_o in the circuit of Fig. 4.8.

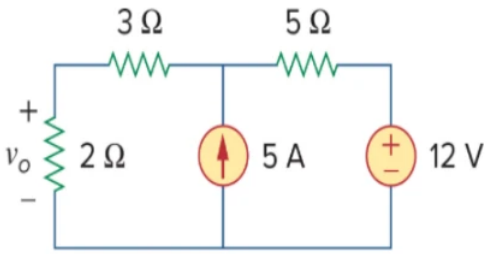


Figure 4.8
For Practice Prob. 4.3.

Answer: 7.4 V.

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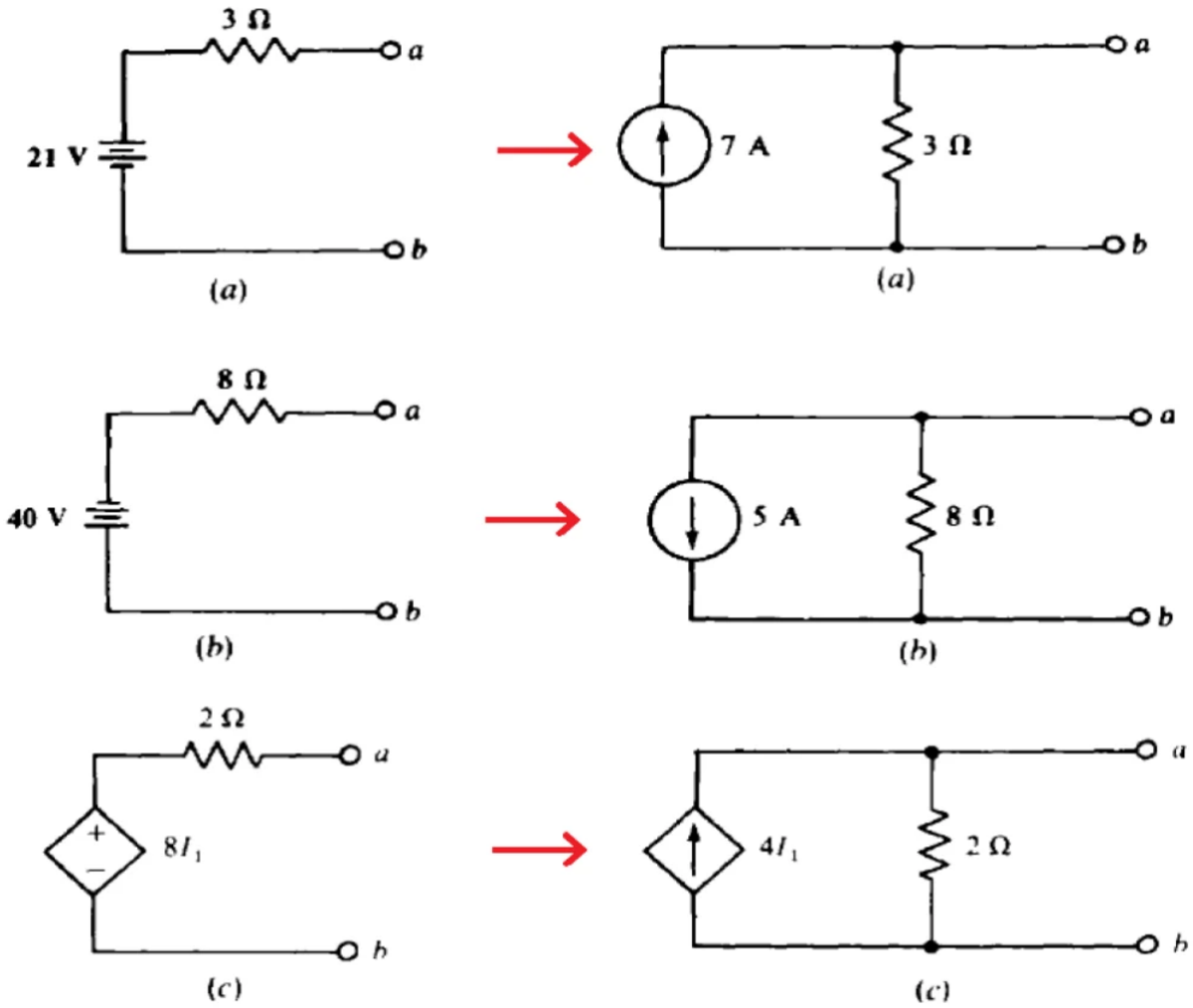
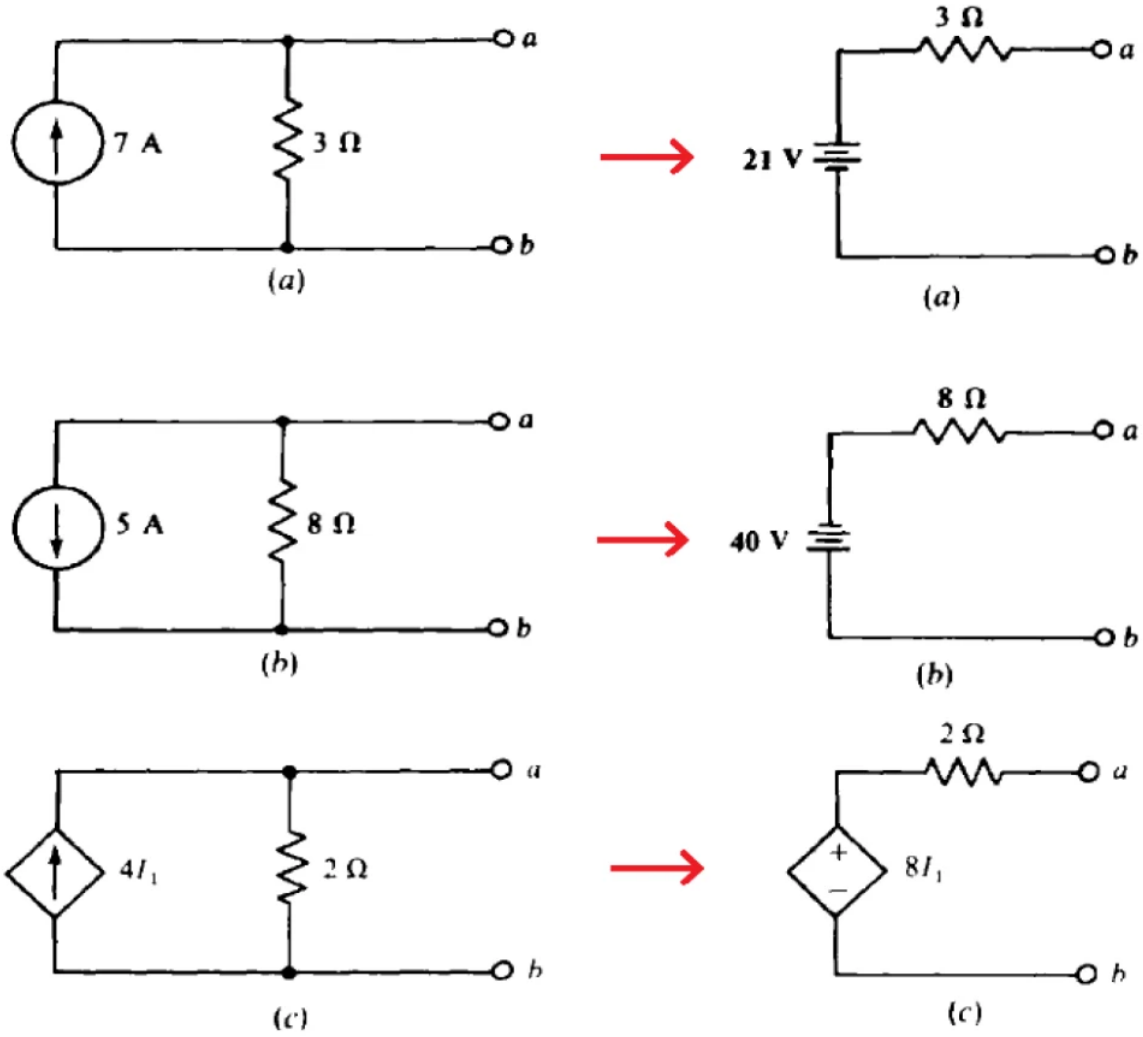


© McGraw Hill



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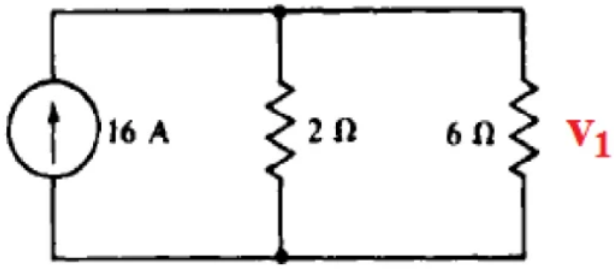
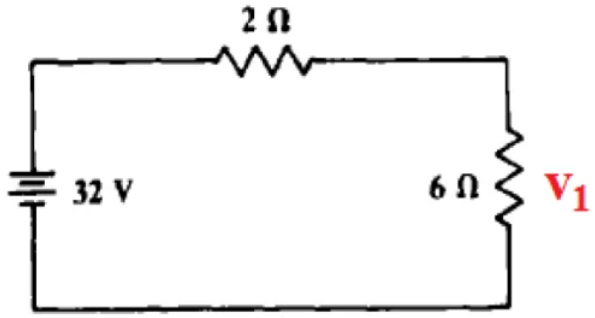
We are interested to find v1, which circuit looks easier to solve ?



(a)



(b)

Circuit (a) can be transformed as circuit (b)

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Example 4.6

Use source transformation to find v_o in the circuit of Fig. 4.17.

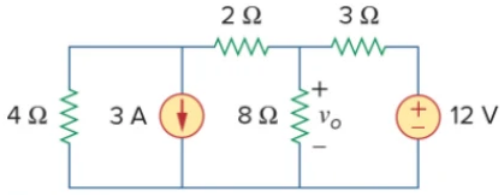

Figure 4.17
For Example 4.6.

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Example 4.7

Find v_x in Fig. 4.20 using source transformation.

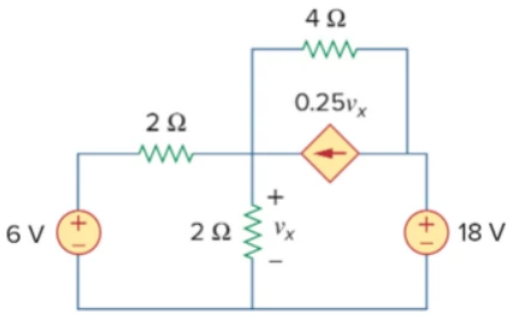

Figure 4.20
For Example 4.7.

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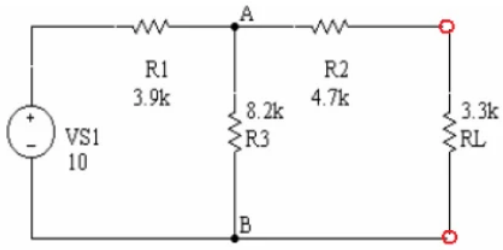
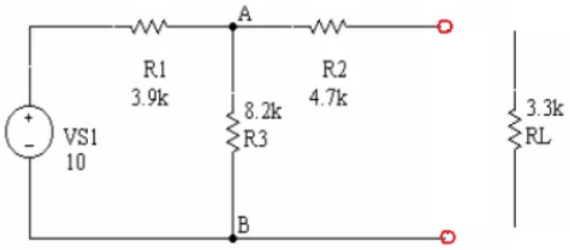
Steps for Thevenin's Equivalent

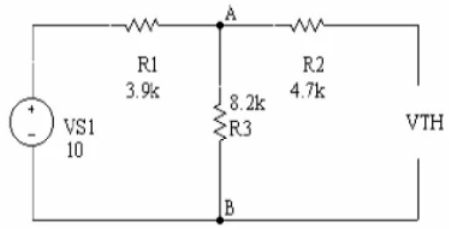
1. **Identify the Load resistance**



2. **Separate the load resistance**



3. **Calculate the open circuit voltage V_{TH}**



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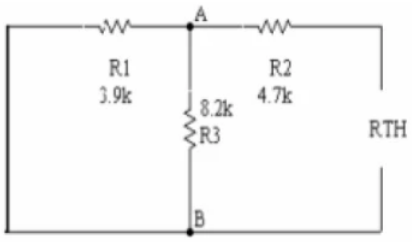
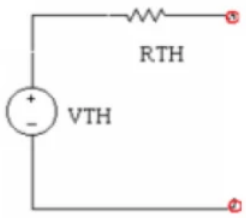
Steps for Thevenin's Equivalent ….

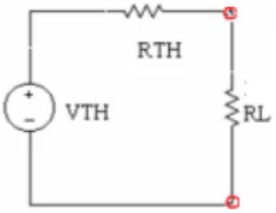
4. Replace all current sources with an open
   Replace voltage sources with a short
   Calculate R_{TH}



5. Draw Thevenin equivalent circuit by putting value of V_{TH} and R_{TH}



6. Connect the load R_L



Now we can easily calculate the load current & voltage etc.

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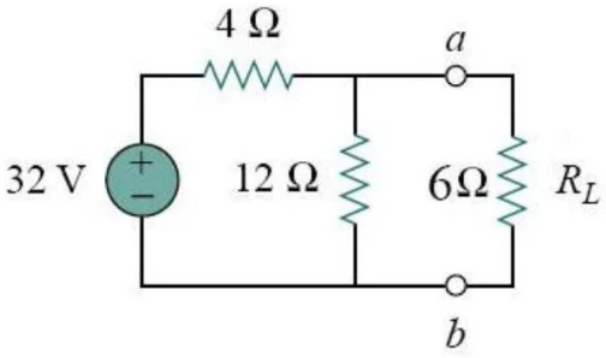
EXAMPLE

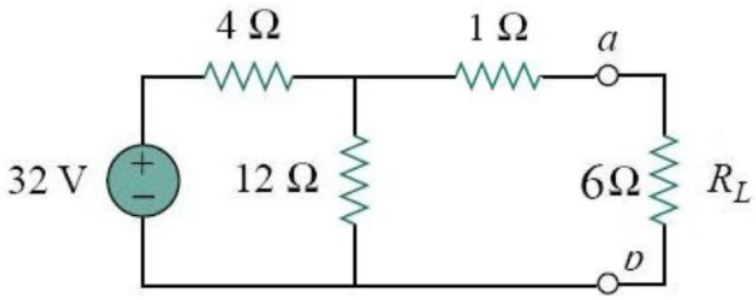
Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a-b. Then find the current through R_L



© McGraw Hill

Example:

Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a-b. Then find the current through R_L



© McGraw Hill

Example 4.8

Find the Thevenin equivalent circuit of the circuit shown in Fig. 4.27, to the left of the terminals *a-b*. Then find the current through $R_L = 6 \quad \Omega$.

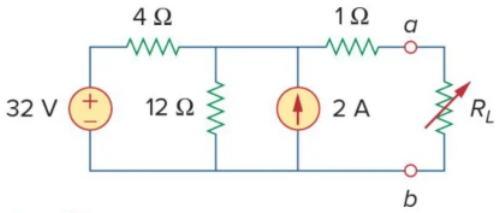


Figure 4.27
For Example 4.8.

Practice Problem 4.8

Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4.30. Then find I.

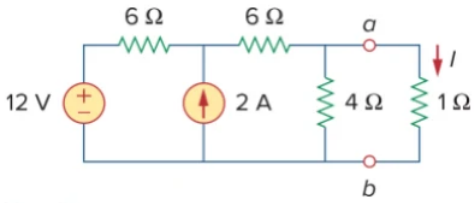


Figure 4.30
For Practice Prob. 4.8.

Answer: $V_{Th} = 6$ V, $R_{Th} = 3\ \Omega$, $I = 1.5$ A.

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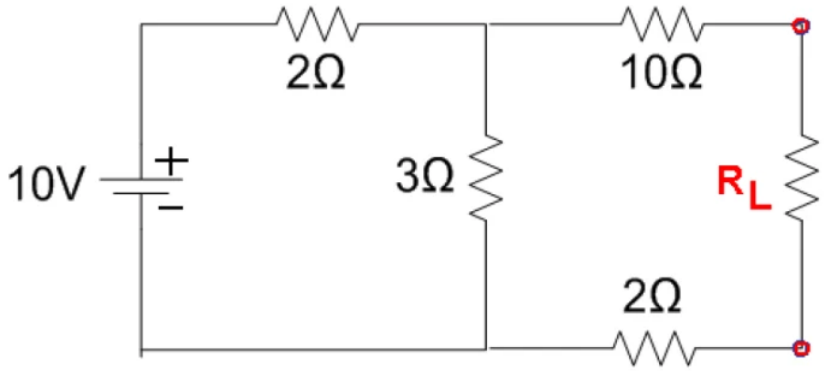
Example

Find Norton's equivalent circuit and current thru R_L, if $R_L = 1\Omega$



© McGraw Hill

Example 4.11

Find the Norton equivalent circuit of the circuit in Fig. 4.39 at terminals *a-b*.

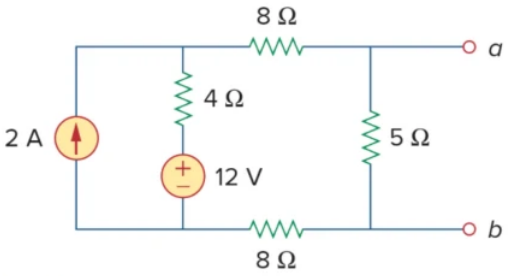


Figure 4.39
For Example 4.11.

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Practice Problem 4.11

Find the Norton equivalent circuit for the circuit in Fig. 4.42, at terminals *a-b*.

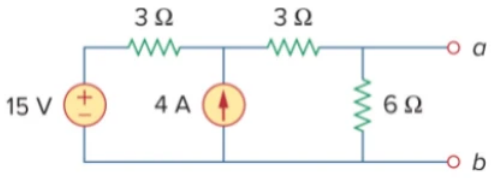

Figure 4.42
For Practice Prob. 4.11.

Answer: $R_N = 3\ \Omega, I_N = 4.5\ \text{A}$.

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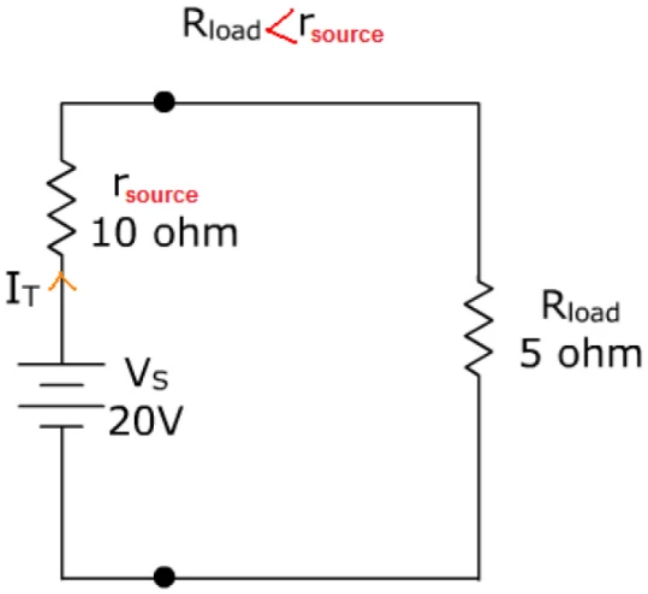
Let's look at some examples.

Example 1



$$R_T = r_{source} + R_{load}$$

$$R_T = 10\ \Omega + 5\ \Omega$$

$$R_T = 15\ \Omega$$

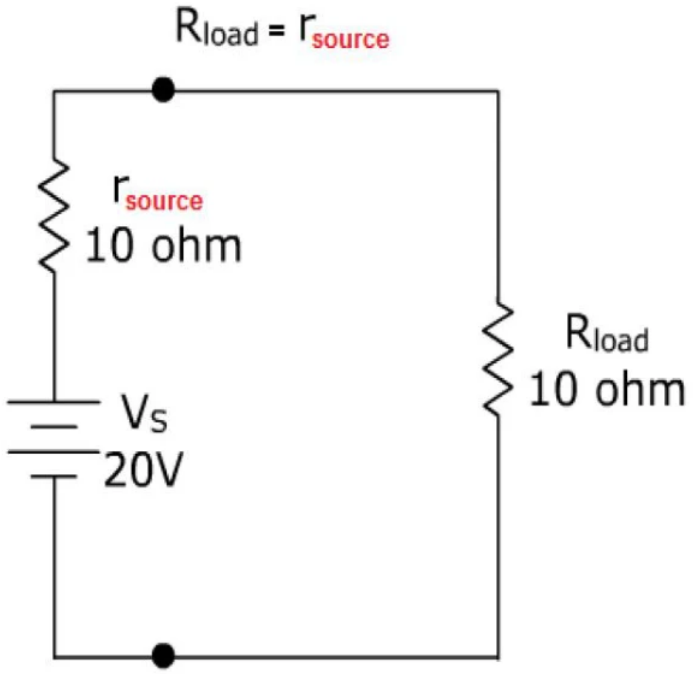
$$I_T = \frac{V_S}{R_T} = \frac{20V}{15\Omega} = 1.33\ A$$

$$P_L = I^2 R_{load} = (1.33\ A)^2\ (5\ \Omega)$$

$$\mathbf{P_L = 8.89\ W}$$

© McGraw Hill

Example 2



$$R_T = r_{source} + R_{load}$$

$$R_T = 10\ \Omega + 10\ \Omega$$

$$R_T = 20\ \Omega$$

$$I_T = \frac{V_S}{R_T} = \frac{20V}{20\Omega} = 1.00\ A$$

$$P_L = I^2 R_{load} = (1.00A)^2\ (10\Omega)$$

$$\mathbf{P_L = 10\ W}$$

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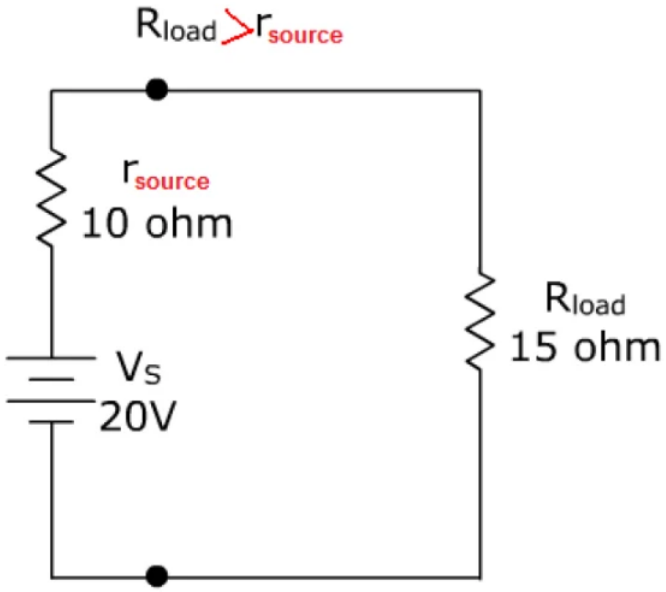

$$R_T = r_{source} + R_{load}$$

$$R_T = 10\ \Omega + 15\ \Omega$$

$$R_T = 25\ \Omega$$

$$I_T = \frac{V_S}{R_T} = \frac{20V}{25\Omega} = 0.80\ A$$

$$P_L = I^2 R_{load} = (0.80A)^2\ (15\Omega)$$

$$\mathbf{P_L = 9.6\ W}$$

© McGraw Hill

Example 4.13

Find the value of R_L for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.

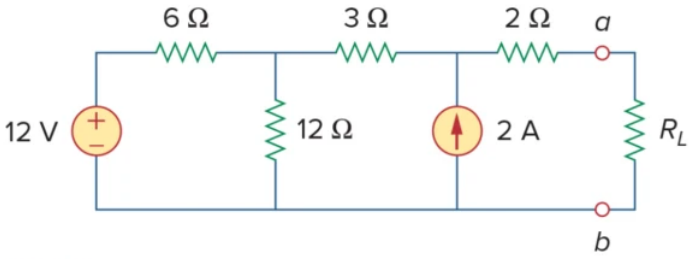

Figure 4.50
For Example 4.13.

Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Problemset 4-Chapter 4- Circuit Theorems

Name: _____ Marks: _____

Section 4.3Superposition

1. **Chapter 4, Problem 11.** Use the superposition principle to find io and vo in the circuit of **Fig. 4.79**.

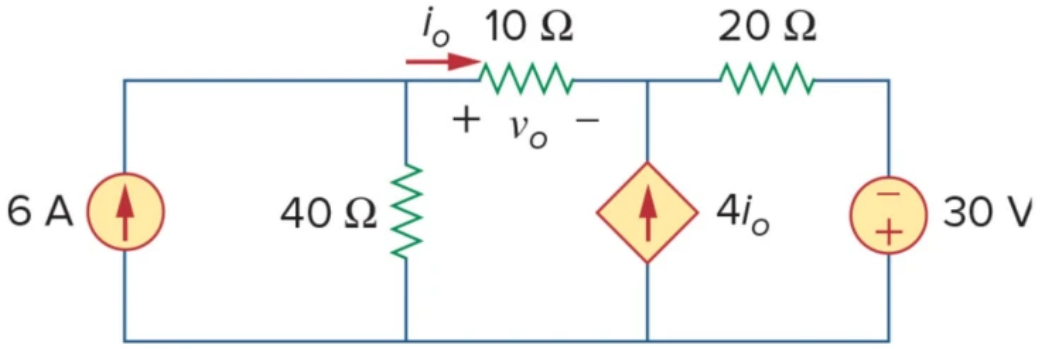

Figure 4.79 For **Prob. 4.11**.

2. **Chapter 4, Problem 12.** Determine Vo in the circuit of **Fig. 4.80** using the superposition principle.

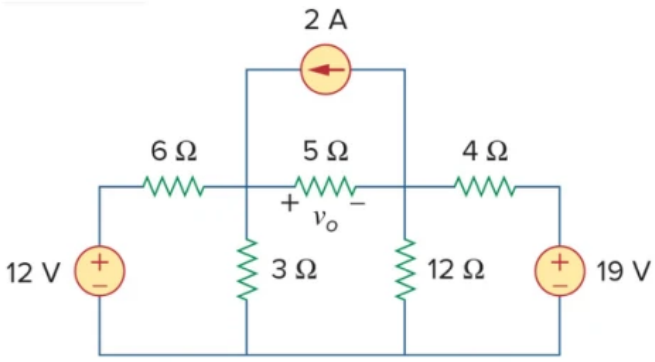

Figure 4.80 For **Prob. 4.12**.

3. Chapter 4, Problem 16. Given the circuit in **Fig. 4.84**, use superposition to obtain io.

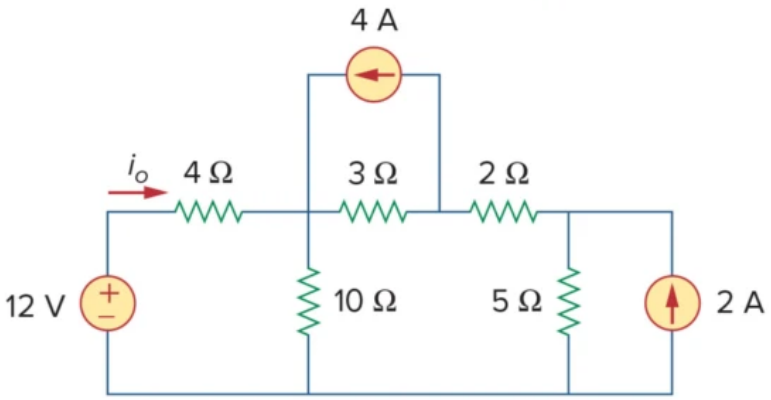

Figure 4.84 For **Prob. 4.16**.

Sections 4.5 and 4.6Thevenin's and Norton's Theorems

4. Chapter 4, Problem 33. Determine the Thevenin equivalent circuit, shown in **Fig. 4.101**, as seen by the 5-ohm resistor. Then calculate the current flowing through the 5-ohm resistor.

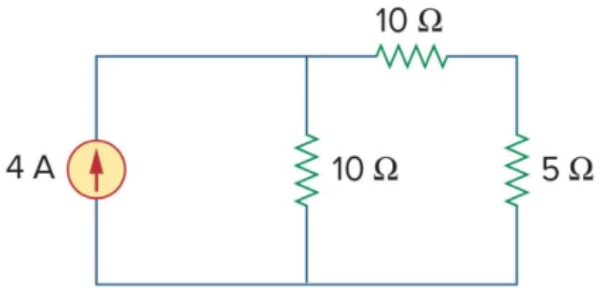

Figure 4.101 For **Prob. 4.33**.

5. Chapter 4, Problem 37. Find the Norton equivalent with respect to terminals *a-b* in the circuit shown in **Fig. 4.104**.

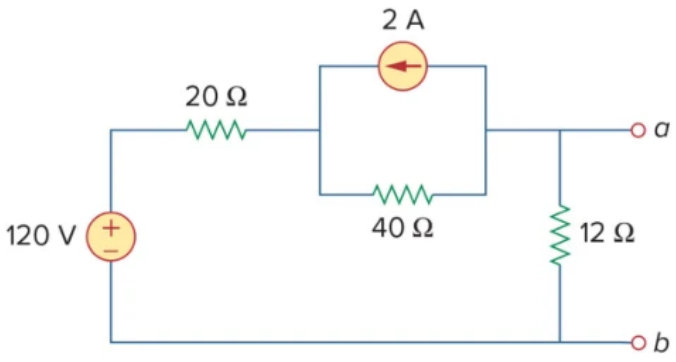

Figure 4.104 For **Prob. 4.37**.

6. **Chapter 4, Problem 36.** Solve for the current i in the circuit of **Fig. 4.103** using Thevenin's theorem. (*Hint:* Find the Thevenin equivalent seen by the 40Ω resistor.)

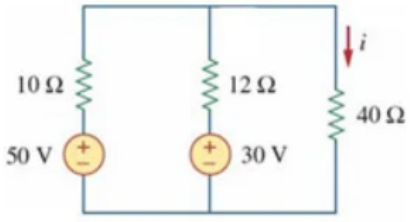

Figure 4.103 For **Prob. 4.36**.

7. **Chapter 4, Problem 38.** Apply Thevenin's theorem to find Vo in the circuit of **Fig. 4.105**.

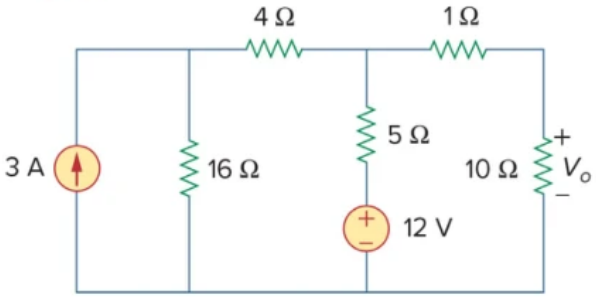

Figure 4.105 For **Prob. 4.38**.

8. Chapter 4, Problem 39. Obtain the Thevenin equivalent at terminals *a-b* of the circuit shown in **Fig. 4.106**.

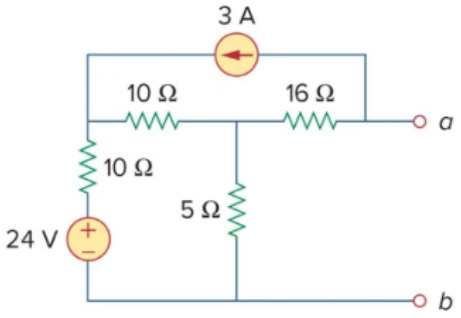

Figure 4.106 For **Prob. 4.39**.

9. **Chapter 4, Problem 41.** Find the Thevenin and Norton equivalents at terminals *a*-*b* of the circuit shown in **Fig. 4.108**.

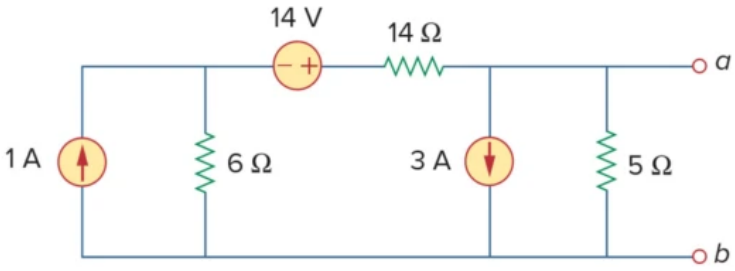

Figure 4.108 For **Prob. 4.41**.

Section 4.8Maximum Power Transfer

10. Chapter 4, Problem 66. Find the maximum power that can be delivered to the resistor R in the circuit of **Fig. 4.132**.

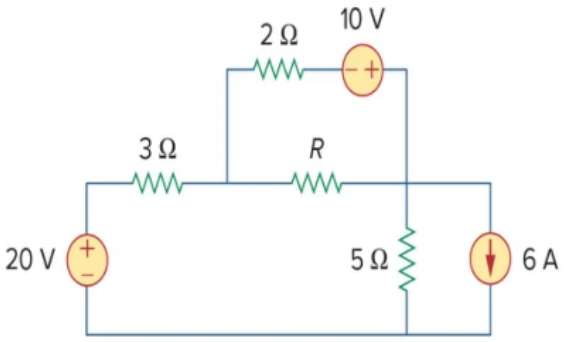

Figure 4.132 For **Prob. 4.66**.

EXAMPLE 6.1

(a) Calculate the charge stored on a 3-pF capacitor with 20 V across it.
(b) Find the energy stored in the capacitor.

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EXAMPLE 6.2

The voltage across a 5-μF capacitor is

$$v(t) = 10\cos 6000t \text{ V}$$

Calculate the current through it.

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EXAMPLE 6.3

Determine the voltage across a 2-μF capacitor if the current through it is

$$i(t) = 6e^{-3000t} \text{ mA}$$

Assume that the initial capacitor voltage is zero.

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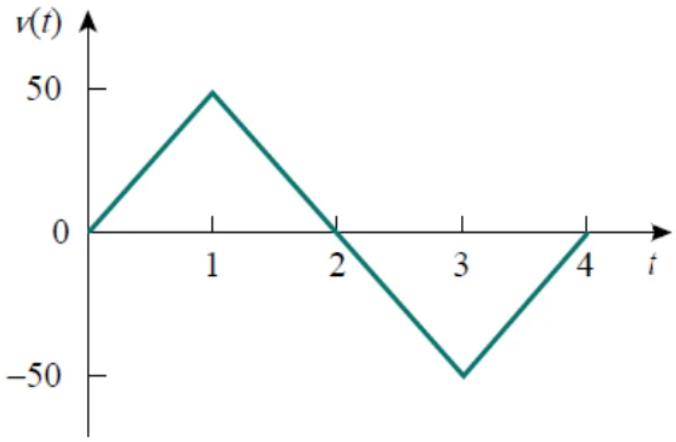
EXAMPLE 6.4

Determine the current through a 200-μF capacitor whose voltage is shown in Fig. 6.9.



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PRACTICE PROBLEM 6.4

An initially uncharged 1-mF capacitor has the current shown in Fig. 6.11 across it. Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.

Answer: 100 mV, 400 mV.

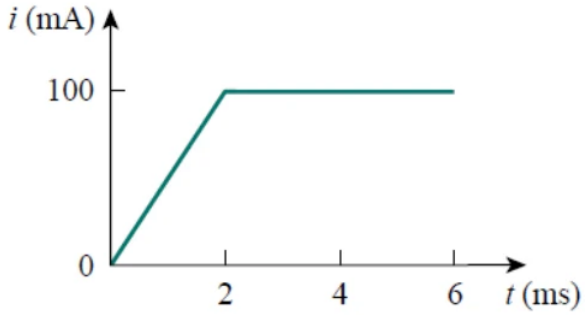

Figure 6.11 For Practice Prob. 6.4.

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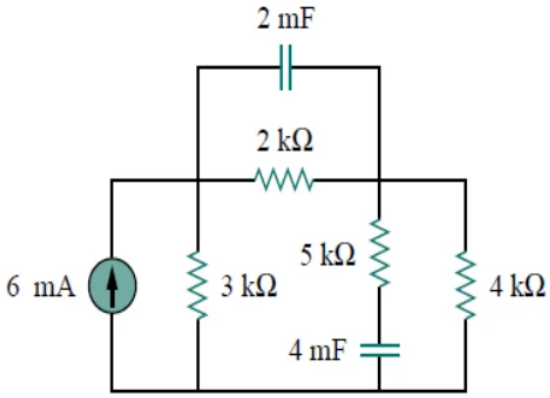
EXAMPLE 6.5

Obtain the energy stored in each capacitor in Fig. 6.12(a) under dc conditions.



(a)

$$w = \frac{1}{2} C v^2$$

Figure 6.12 For Example 6.5.

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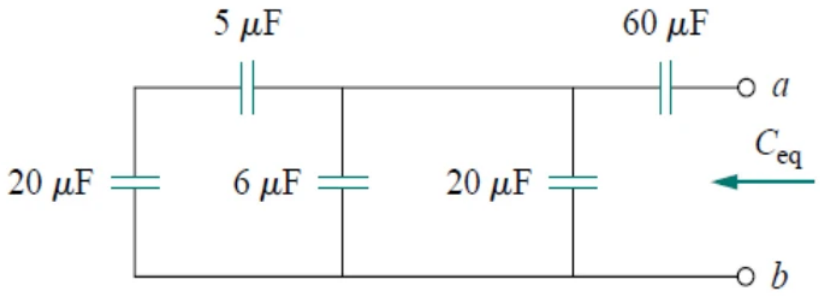
EXAMPLE 6.6

Find the equivalent capacitance seen between terminals a and b of the circuit in Fig. 6.16.



Ans: 20 μF

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Example 6.8

The current through a 0.1-H inductor is $i(t) = 10te^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

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Example 6.9

Find the current through a 5-H inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also, find the energy stored at $t = 5$ s. Assume $i(v) > 0$.

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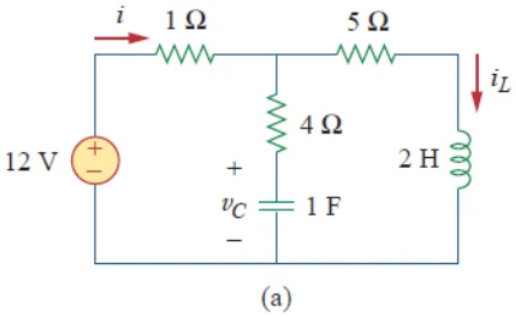
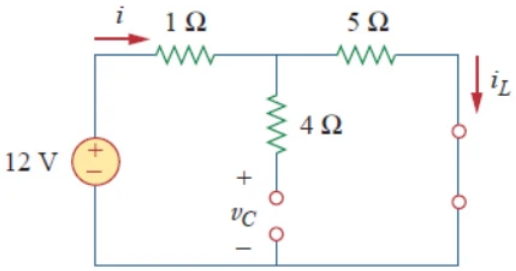
Example 6.10

Consider the circuit in Fig. 6.27(a). Under dc conditions, find: (a) i, v_C, and i_L, (b) the energy stored in the capacitor and inductor.



(a)



(b)

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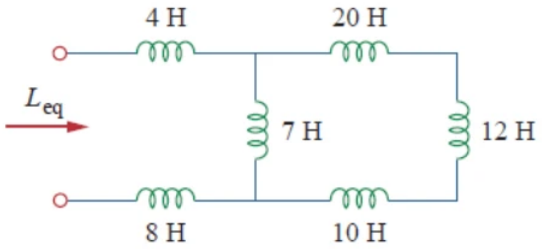
Example 6.11

Find the equivalent inductance of the circuit shown in Fig.



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Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

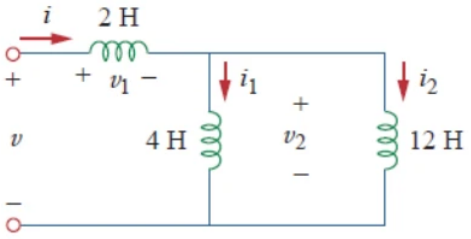

Figure 6.33
For Example 6.12.

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Example 6.12

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

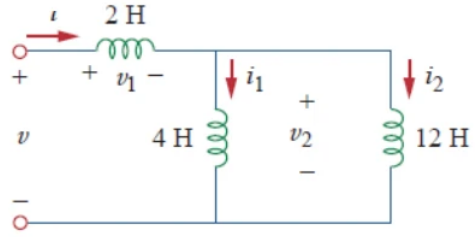

Figure 6.33
For Example 6.12.

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Example 7.1

In Fig. 7.5, let $v_C(0) = 15$ V. Find v_C, v_x, and i_x for $t > 0$.

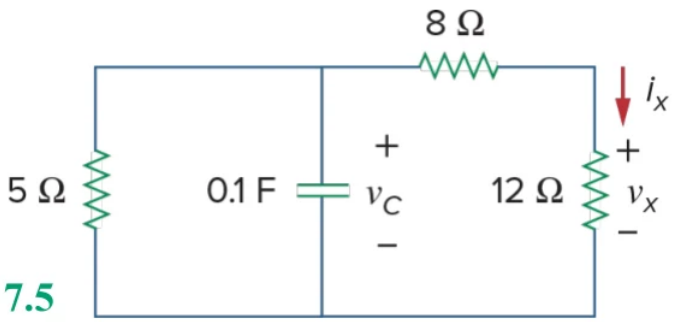

Figure 7.5
For Example 7.1.

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Example 7.2

The switch in the circuit in Fig. 7.8 has been closed for a long time, and it is opened at $t = 0$. Find $v(t)$ for $t \geq 0$. Calculate the initial energy stored in the capacitor.

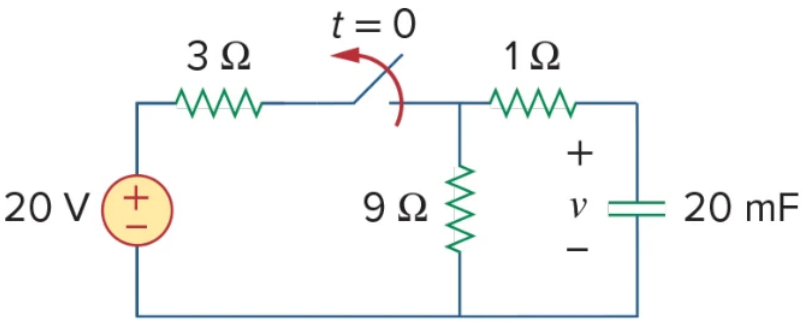

Figure 7.8
For Example 7.2.

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Example 7.4

The switch in the circuit of Fig. 7.16 has been closed for a long time. At $t = 0$, the switch is opened. Calculate $i(t)$ for $t > 0$.

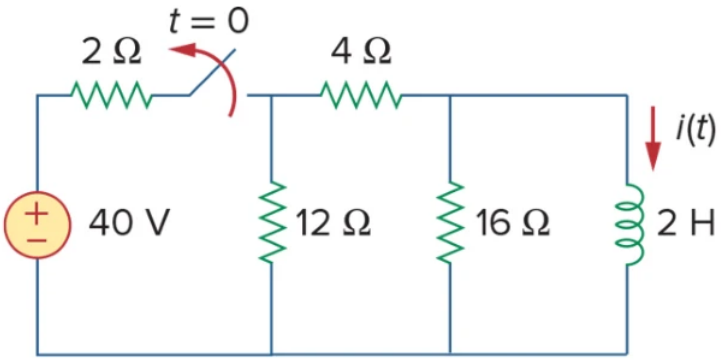

Figure 7.16 For **Example 7.4**.

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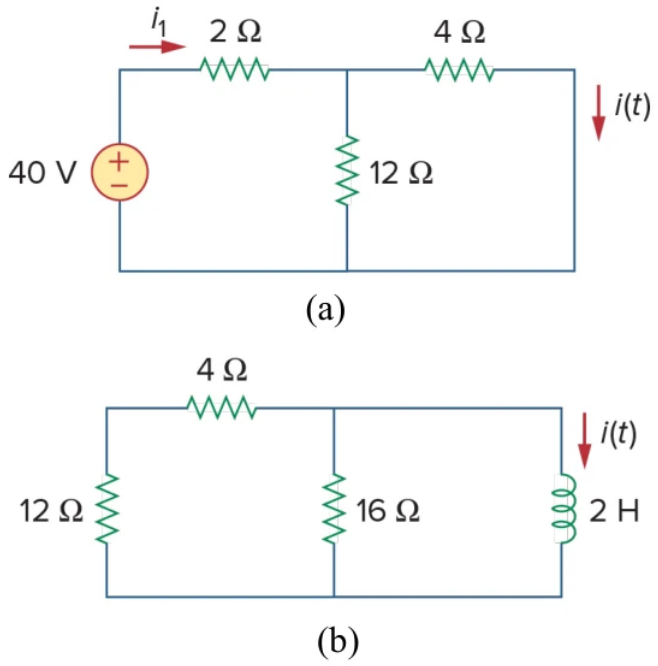

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Example 7.10

The switch in Fig. 7.43 has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ and 4 s.

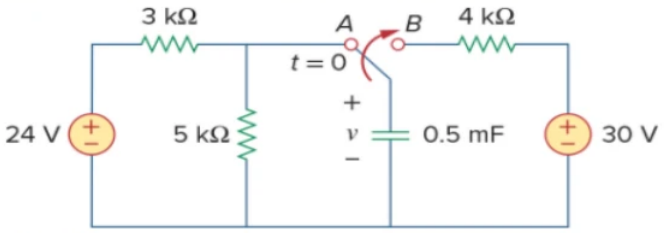

Figure 7.43
For Example 7.10.

Solution:

For $t < 0$, the switch is at position A. The capacitor acts like an open circuit to dc, but v is the same as the voltage across the 5-kΩ resistor. Hence, the voltage across the capacitor just before $t = 0$ is obtained by voltage division as

$$v(0^-) = \frac{5}{5+3}(24) = 15 \text{ V}$$

Using the fact that the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = v(0^+) = 15 \text{ V}$$

For $t > 0$, the switch is in position B. The Thevenin resistance connected to the capacitor is $R_{\text{Th}} = 4$ kΩ, and the time constant is

$$\tau = R_{\text{Th}} C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s}$$

Since the capacitor acts like an open circuit to dc at steady state, $v(\infty) = 30$ V. Thus,

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 30 + (15 - 30)e^{-t/2} = \left(30 - 15e^{-0.5t}\right) \text{ V} \end{aligned}$$

At $t = 1$,

$$v(1) = 30 - 15e^{-0.5} = 20.9 \text{ V}$$

At $t = 4$,

$$v(4) = 30 - 15e^{-2} = 27.97 \text{ V}$$

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Example 7.11

In Fig. 7.45, the switch has been closed for a long time and is opened at $t = 0$. Find i and v for all time.

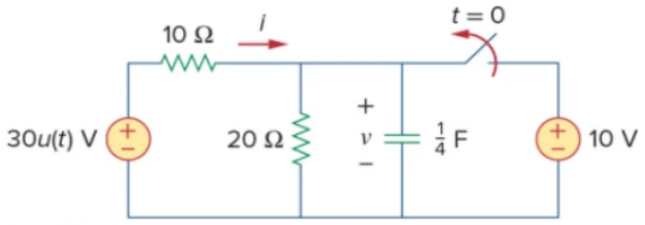

Figure 7.45
For Example 7.11.

Solution:

The resistor current i can be discontinuous at $t = 0$, while the capacitor voltage v cannot. Hence, it is always better to find v and then obtain i from v.

By definition of the unit step function,

$$30u(t) = \begin{cases} 0, & t < 0 \\ 30, & t > 0 \end{cases}$$

For $t < 0$, the switch is closed and $30u(t) = 0$, so that the $30u(t)$ voltage source is replaced by a short circuit and should be regarded as contributing nothing to v. Since the switch has been closed for a long time, the capacitor voltage has reached steady state and the capacitor acts like an open circuit. Hence, the circuit becomes that shown in Fig. 7.46(a) for $t < 0$. From this circuit we obtain

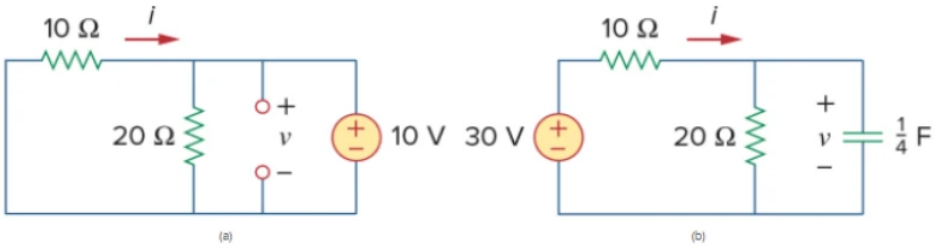

Figure 7.46
Solution of Example 7.11: (a) for $t < 0$, (b) for $t > 0$.

$$v = 10 \text{ V}, \quad i = -\frac{v}{10} = -1 \text{ A}$$

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Since the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = 10 \text{ V}$$

For $t > 0$, the switch is opened and the 10-V voltage source is disconnected from the circuit. The $30u(t)$ voltage source is now operative, so the circuit becomes that shown in Fig. 7.46(b). After a long time, the circuit reaches steady state and the capacitor acts like an open circuit again. We obtain $v(\infty)$ by using voltage division, writing

$$v(\infty) = \frac{20}{20 + 10}(30) = 20 \text{ V}$$

The Thevenin resistance at the capacitor terminals is

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$$R_{\text{Th}} = 10 \| 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega$$

and the time constant is

$$\tau = R_{\text{Th}} C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3} s$$

Thus,

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = \left(20 - 10e^{-0.6t}\right) \text{ V} \end{aligned}$$

To obtain i, we notice from Fig. 7.46(b) that i is the sum of the currents through the 20-Ω resistor and the capacitor; that is,

$$\begin{aligned} i &= \frac{v}{20} + C\frac{dv}{dt} \\ &= 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = \left(1 + e^{-0.6t}\right) \text{ A} \end{aligned}$$

Notice from Fig. 7.46(b) that $v + 10i = 30$ is satisfied, as expected. Hence,

$$v = \begin{cases} 10 \text{ V}, & t < 0 \\ \left(20 - 10e^{-0.6t}\right) \text{ V}, & t \geq 0 \end{cases}$$

$$i = \begin{cases} -1 \text{ A}, & t < 0 \\ \left(1 + e^{-0.6t}\right) \text{ A}, & t > 0 \end{cases}$$

Notice that the capacitor voltage is continuous while the resistor current is not.

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Example 7.12

Find $i(t)$ in the circuit of Fig. 7.51 for $t > 0$. Assume that the switch has been closed for a long time.

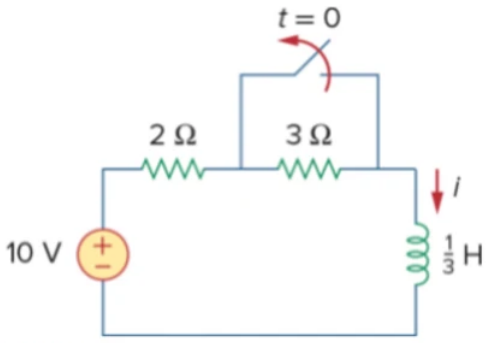

Figure 7.51
For Example 7.12.

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ECET 250E-Linear Circuit 1
Problemset 5-Chapter 6- Capacitors and Inductors

Name: _____ Marks: _____

1. **6.16** The equivalent capacitance at terminals *a*-*b* in the circuit of **Fig. 6.50** is 30 μF. Calculate the value of *C*.

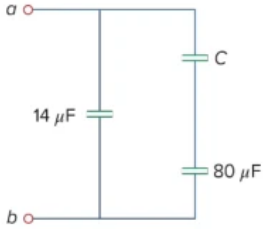


Figure 6.50 For **Prob. 6.16**.

2. **6.17** Determine the equivalent capacitance for each of the circuits of **Fig. 6.51**.

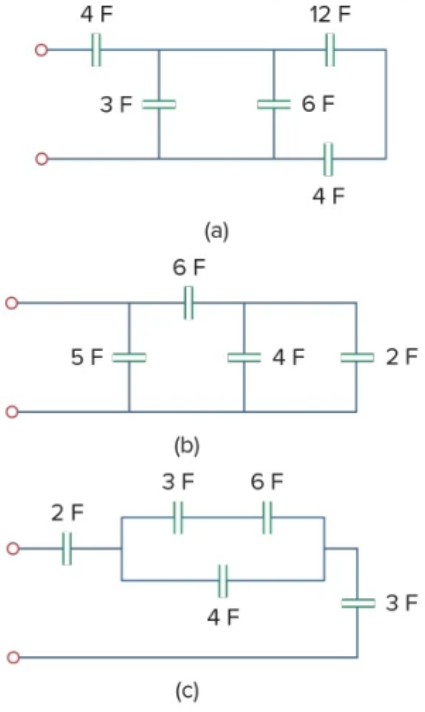


Figure 6.51 For **Prob. 6.17**.

3. **6.18** Find Ceq in the circuit of **Fig. 6.52** if all capacitors are 4 μF.

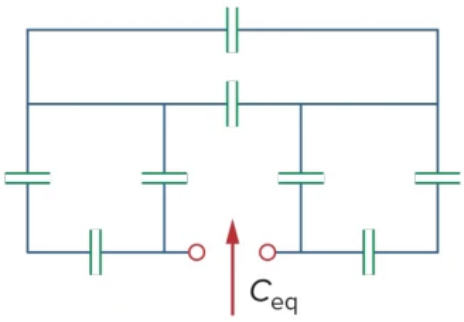


Figure 6.52 For **Prob. 6.18**.

4. **6.26** Three capacitors, C1=5 μF, C2=10 μF, and C3=20 μF, are connected in parallel across a 150-V source. Determine:
   1. (a)the total capacitance,
   2. (b)the charge on each capacitor,
   3. (c)the total energy stored in the parallel combination

5. **6.32** In the circuit in **Fig. 6.64**, let is=$50e^{-2t}$ mA and $v_1(0)$=50 V, $v_2(0)$=20 V. Determine: (a) $v_1(t)$ and $v_2(t)$, (b) the energy in each capacitor at t=0.5 s.

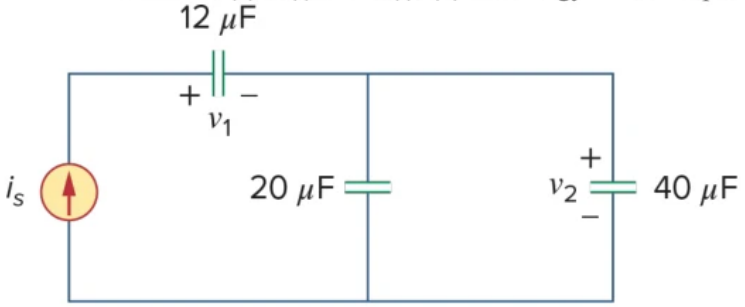


Figure 6.64 For **Prob. 6.32**.

6. **6.40** The current through a 5-mH inductor is shown in **Fig. 6.66**. Determine the voltage across the inductor at t=1, 3, and 5 ms.

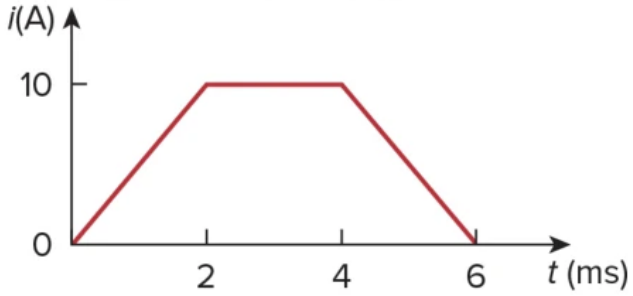


Figure 6.66 For **Prob. 6.40**.

7. **6.43** The current in an 80-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

8. **6. 51** Determine Leq at terminals *a-b* of the circuit in **Fig. 6.73**.

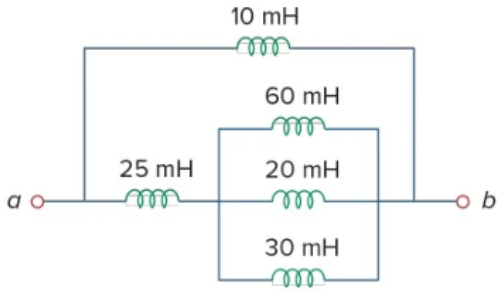


Figure 6.73 For **Prob. 6.51**.

9. **6.53** Find Leq at the terminals of the circuit in **Fig. 6.75**.

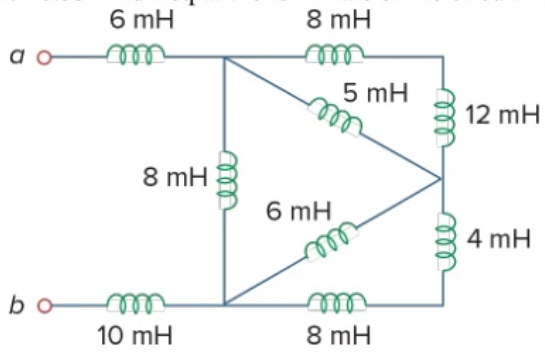


Figure 6.75 For **Prob. 6.53**.

10. 6.60 In the circuit of **Fig. 6.82**, io(0)=2 A. Determine io(t) and vo(t) for t>0.

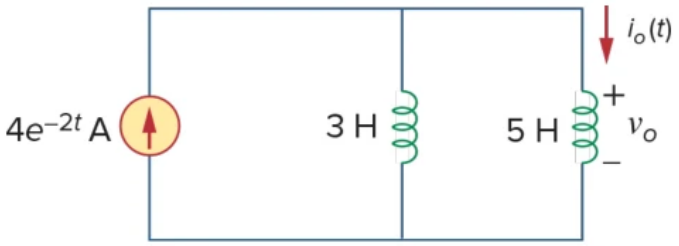


Figure 6.82 For **Prob. 6.60**.

Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Problemset 6-Chapter 7- First Order RC-RL Circuits

Name: _____ Marks: _____

1. **7.2** Find the time constant for the *RC* circuit in Fig. 7.82.

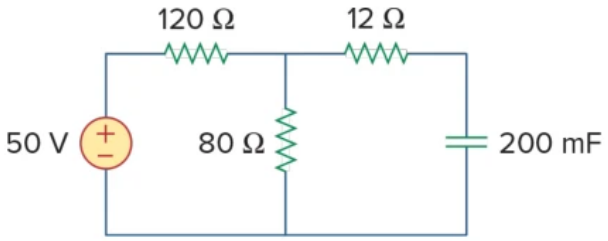


Figure 7.82 For Prob. 7.2.

2. **7.6** The switch in Fig. 7.86 has been closed for a long time, and it opens at t=0. Find v(t) for t≥0.

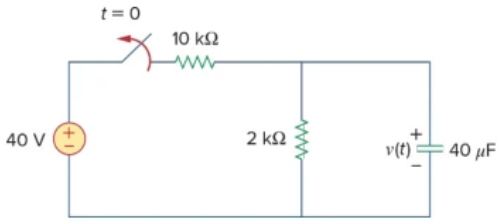


Figure 7.86 For Prob. 7.6.

3. 7.9 The switch in Fig. 7.89 opens at t=0. Find vo for t>0.

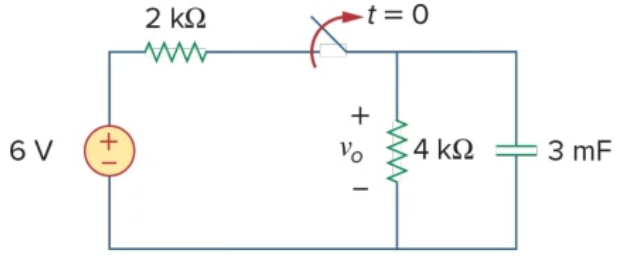


Figure 7.89 For Prob. 7.9.

4. 7.11 For the circuit in Fig. 7.91, find io for t>0.

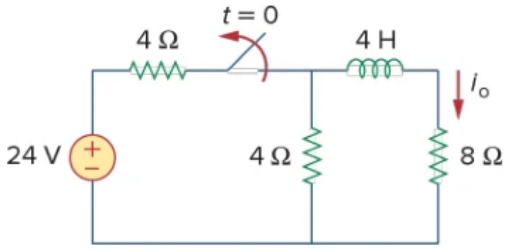


Figure 7.91 For Prob. 7.11.

5. 7.13. In the circuit of Fig. 7.93,

$$v(t) = 80e^{-10^3 t}\ \text{V}, \quad t > 0$$
$$i(t) = 5e^{-10^3 t}\ \text{mA}, \quad t > 0$$

(a)Find R, L, and τ.
(b)Calculate the energy dissipated in the resistance for 0<t<0.5 ms.

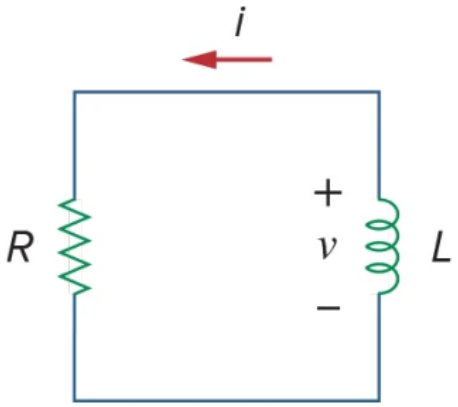


Figure 7.93 For Prob. 7.13.

6. **7.14** Calculate the time constant of the circuit in Fig. 7.94.

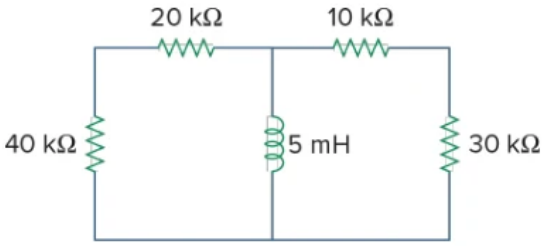


Figure 7.94 For Prob. 7.14.

7. **7.39** Calculate the capacitor voltage for t<0 and t>0 for each of the circuits in Fig. 7.106.

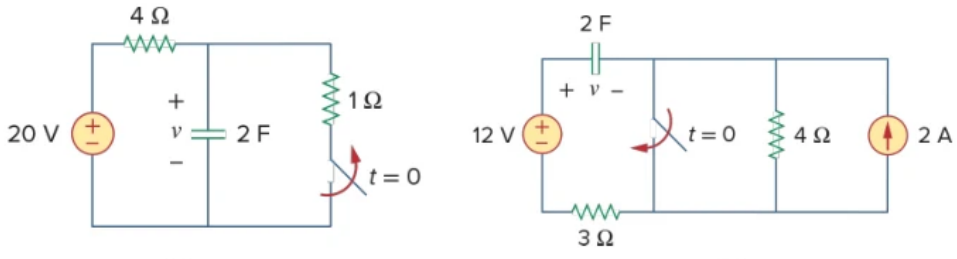
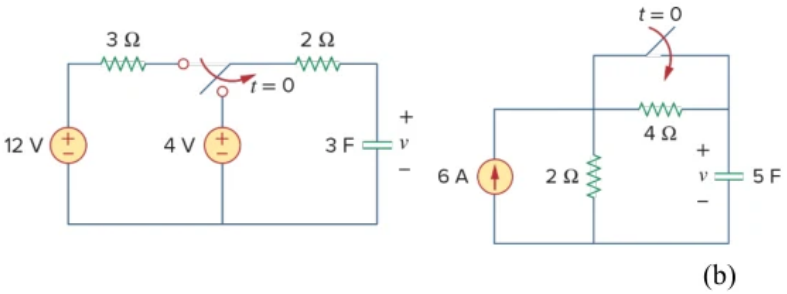


Figure 7.106 For Prob. 7.39.

8. 7.40 Find the capacitor voltage for t<0 and t>0 for each of the circuits in Fig. 7.107.



(a) (b)

Figure 7.107 For Prob. 7.40.

9. 7.42 (a)If the switch in Fig. 7.109 has been open for a long time and is closed at t=0, find $v_o(t)$.

(b)Suppose that the switch has been closed for a long time and is opened at t=0. Find $v_o(t)$.

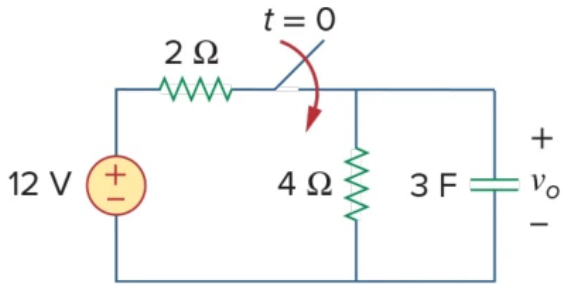


Figure 7.109 For Prob. 7.42.

Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Quiz 1-Chapter 1&2- Basic Concepts & Laws
Solution

Name: _____

Total Marks: _____ / 30	**Percentage:** _____ %

1. The dependent source in figure below is:

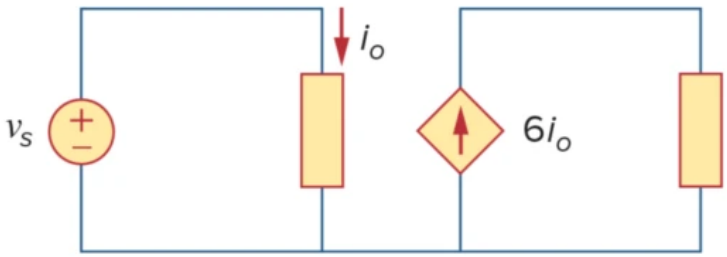
(a)voltage-controlled current source

(b)voltage-controlled voltage source

(c)current-controlled voltage source

(d)current-controlled current source

Answer: *d*



2. Determine the current flowing through an element if the charge flow is given by $\boldsymbol{q(t) = (9e^{-3t} - 2e^{-6t})\ nC}$

[Hint: i = dq/dt]

$i = dq/dt = d/dt\ (9e^{-3t} - 2e^{-6t}) = (-27e^{-3t}) + (12e^{-6t}) =$

$= (-27e^{-3t} + 12e^{-6t})\ nA$ *Marks (2): 1 for each correct derivative term.*

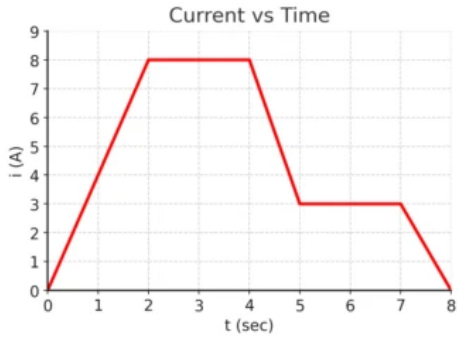
3. Determine the **total charge** transferred over the time interval of **$0 \leq t \leq 6s$** when $\boldsymbol{i(t) = \frac{1}{3}t}$ **A**.

Solution:

$$q = \int idt = \int_0^6 \frac{1}{3} t dt = \frac{1}{3}\left(\frac{t^2}{2}\right)\Big|_0^6 = \frac{1}{3}\left(\frac{6^2}{2} - \frac{0^2}{2}\right) = \frac{1}{3} * \frac{36}{2} = \frac{1}{3} * 18 = 6 \text{ C}$$

Marks (2): 1 for the correct integration, 1 for applying the limits and the final answer (6 C).

4. The current flowing past a point in a device is shown in Fig. below. Calculate the total charge through the point



Solution:

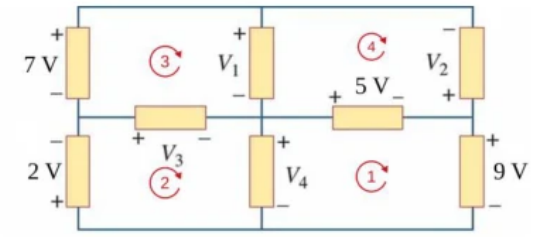
$$q = \int \text{idt} = A1 + A2 + A3 + A4 + A5$$

$$= \left(\frac{1}{2} \times 2 \times 8\right) + (2 \times 8) + \left(\frac{1}{2} \times 1 \times 5\right) + (3 \times 3)$$

$$+ \left(\frac{1}{2} \times 1 \times 3\right) = 8 + 16 + 2.5 + 9 + 1.5 = \underline{37\ C}$$

Marks (4): 3 for the areas A1 to A5, 1 for the total charge with units (37 C).

5. Given the circuit in Fig. below, use KVL to find the branch voltages V_1 to V_4.



For mesh 1,

$$-V_4 + 5 + 9 = 0$$

$$V_4 = 14 \text{ V}$$

For mesh 2,

$$+2 + V_3 + V_4 = 0$$

$$V_3 = -2 - 14 = -16 \text{ V}$$

For mesh 3,

$$-7 + V_1 - V_3 = 0$$

$$V_1 = V_3 + 7 = -16 + 7 = -9 \text{ V}$$

For mesh 4,

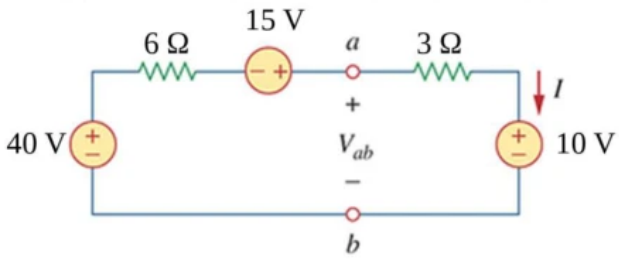
$$-V_1 - V_2 - 5 = 0$$

$$V_2 = -V_1 - 5 = -(-9) - 5 = 9 - 5 = 4 \text{ V}$$

Thus, $V_1 = -9$ V, $V_2 = 4$ V, $V_3 = -16$ V, $V_4 = 14$ V

Marks (4): 1 for each branch voltage V_1 to V_4 with its KVL equation.

6. Find I and V_{ab} in the circuit of Fig. Below.



Applying KVL,

$-40 -15 +10 + I(6+3) = 0$

$9I = 45$ $\quad I = \mathbf{5A}$

$-V_{ab} + 3I + 10 = 0$ $\quad -V_{ab} + (3*5) + 10 = 0$ $\quad V_{ab} = \mathbf{25V}$

Marks (4): 1 for the KVL loop equation, 1 for I, 1 for the V_{ab} equation, 1 for V_{ab}.

7. The total power dissipated in a parallel circuit is equal to the

a. power in the largest resistor.

b. power in the smallest resistor.

c. average of the power in all resistors.

d. sum of the power in all resistors.

answer d

8. KCL is based on the law of conservation of

a. energy.

b. charge.

c. voltage.

d. power.

answer b

9. Which of the following ideal circuit elements dissipates energy

a. inductor.

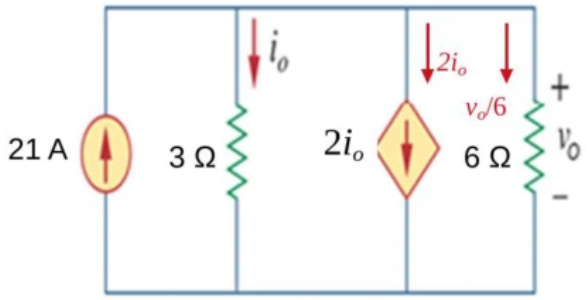
b. capacitor.

c. resistor.

d. none of the above.

answer c

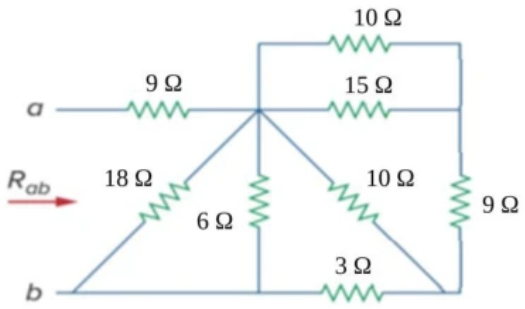
10. Find V_o, I_o in the circuit below:



Applying KCL, $21 - i_o - [2i_o] - [v_o/6] = 0$, but $i_o = v_o/3$
Which leads to: $21 = (v_o/3) + (2v_o/3) + (v_o/6)$ thus, $v_o =$ **18 V** and $i_o =$ **6 A**

Marks (4): 1 for the KCL equation, 1 for the control relation $i_o = v_o/3$, 1 for v_o, 1 for i_o.

11. Find $\mathbf{R_{ab}}$ (Resistor Equal) for the circuit below.



Solution:

Step 1: The top 10 Ω and the middle 15 Ω resistors are in parallel: $10 \parallel 15 =$

$(10 \times 15)/(10+15) = 150/25 = 6\ \Omega$

Step 2: That 6 Ω is in series with the 9 Ω on the right: $6 + 9 = 15\ \Omega$

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Example 8.1

The switch in Fig. 8.2 has been closed for a long time. It is open at $t = 0$. Find: (a) $i(0^+)$, $v(0^+)$, (b) $di(0^+)/dt$, $dv(0^+)/dt$, (c) $i(\infty)$, $v(\infty)$.

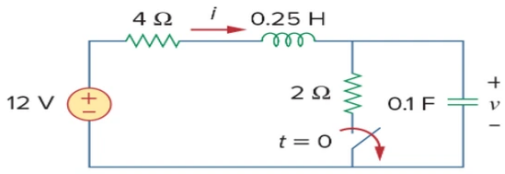


Figure 8.2
For Example 8.1.

Example 8.3

In Fig. 8.8, $R = 40\ \Omega$, $L = 4$ H, and $C = 1/4$ F. Calculate the characteristic roots of the circuit. Is the natural response overdamped, underdamped, or critically damped?

Example 8.4

Find $i(t)$ in the circuit of Fig. 8.10. Assume that the circuit has reached steady state at $t = 0^-$.

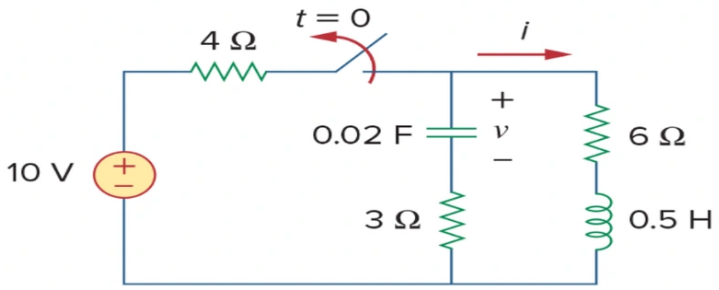


Figure 8.10
For Example 8.4.

Example 8.5

In the parallel circuit of Fig. 8.13, find $v(t)$ for $t > 0$, assuming $v(0) = 5$ V, $i(0) = 0$, $L = 1$ H, and $C = 10$ mF. Consider these cases: $R = 1.923\ \Omega$, $R = 5\ \Omega$, and $R = 6.25\ \Omega$.

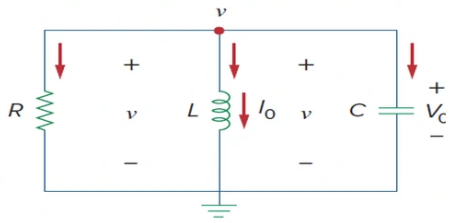


Figure 8.13
A source-free parallel *RLC* circuit.

Example 8.8

In the circuit of Fig. 8.23, find $i(t)$ and $i_R(t)$ for $t > 0$.

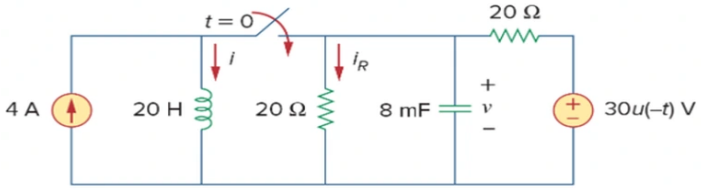


Figure 8.23
For Example 8.8.

Camosun College- Electronics Department

ECET 250E-Linear Circuit 1
Problemset 7-Chapter 8- Second Order RLC Circuits

Name: _____ Marks: _____

Source Free Series RLC Circuits:

1. **Problem 8.13** For the circuit in Fig. 8.68, calculate the value of R needed to have a critically damped response.

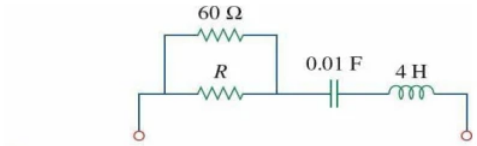


Figure 8.68
For Prob. 8.13.

2. **Problem 8.17** In the circuit of Fig. 8.71, the switch instantaneously moves from position A to B at $t = 0$. Find $v(t)$ for all $t \geq 0$.

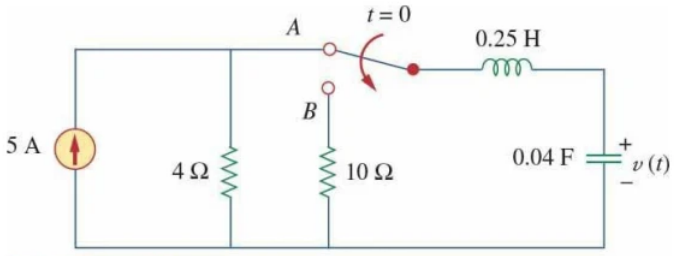


Figure 8.71
For Prob. 8.17.

SOURCE FREE PARALLEL RLC:

3. **Problem 8.23** For the network in Fig. 8.76, what value of C is needed to make the response underdamped with unity damping factor ($\alpha = 1$)?

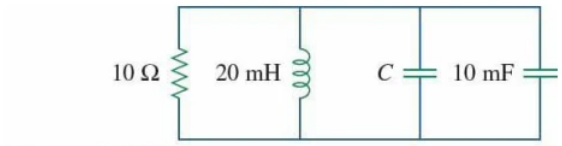


Figure 8.76
For Prob. 8.23.

Step response parallel RLC circuits:

4. Problem 8.47 Find the output voltage $v_o(t)$ in the circuit of Fig. 8.94.

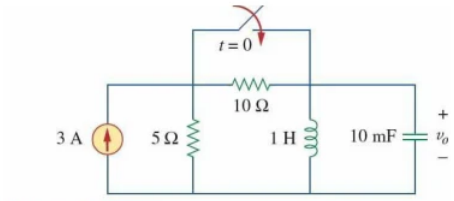


Figure 8.94
For Prob. 8.47.

5. Problem 8.49 Determine $i(t)$ for $t > 0$ in the circuit of Fig. 8.96.

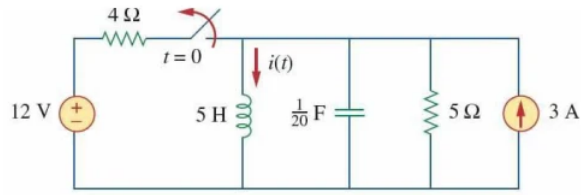


Figure 8.96
For Prob. 8.49.

$\omega_o = 1/\sqrt{0.02x0.05} = 141.42 > \alpha$ (underdamped)

$C_o = C + 10\text{ mF} = 50\text{ mF}$ or $C = \mathbf{40\ mF}$

Step Response of a Series RLC Circuit

4. **Problem 8.33** Find $v(t)$ for $t > 0$ in the circuit of Fig. 8.81.

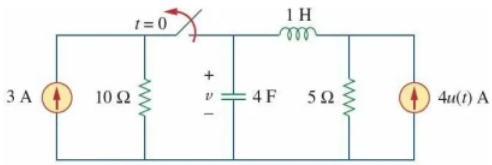
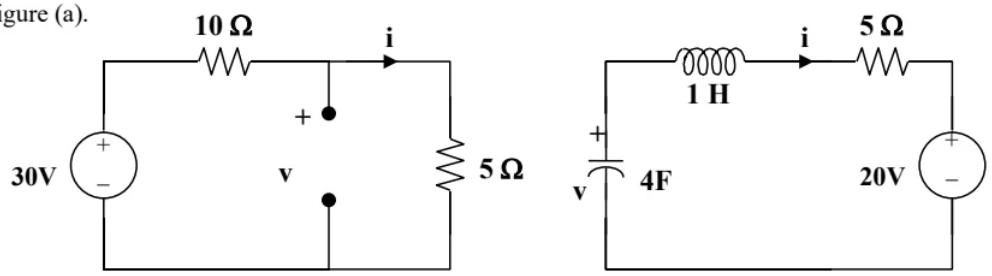


Figure 8.81
For Prob. 8.33.

Chapter 8, Solution 33.

We may transform the current sources to voltage sources. For $t = 0^-$, the equivalent circuit is shown in Figure (a).



(a) **(b)**

$i(0) = 30/15 = 2\text{ A}, \quad v(0) = 5x30/15 = 10\text{ V}$

For $t > 0$, we have a series RLC circuit, shown in (b).

$$\alpha = R/(2L) = 5/2 = 2.5$$

$$\omega_o = 1/\sqrt{LC} = 1/\sqrt{4}$$

$$-\alpha \pm \sqrt{\alpha^2 - \omega_o^2} = -2.5 \pm \sqrt{6.25 - 0.25}$$

$v(t) = V_s + [A_1e^{-4.949t} + A_2e^{-0.0505t}], \quad V_s = 20.$

$v(0) = 10 = 20 + A_1 + A_2$ or

$A_2 = -10 - A_1 \quad (1)$

$i(0) = Cdv(0)/dt$ or $dv(0)/dt = -2/4 = -1/2$

Hence, $-0.5 = -4.949A_1 - 0.0505A_2 \quad (2)$

From (1) and (2), $-0.5 = -4.949A_1 + 0.0505(10 + A_1)$ or

$-4.898A_1 = -0.5 - 0.505 = -1.005$

$A_1 = 0.2052, \quad A_2 = -10.205$

$v(t) = \mathbf{[20 + 0.2052e^{-4.949t} - 10.205e^{-0.05t}]\ V}.$

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